

Behavioral Finance Research Project

Why Markets Aren't Rational, and What You Can Do About It

A project guide covering research design, methodology, five real-world case studies, the psychology behind each one, the milestone-by-milestone decisions made while building it, the real (not simulated) results obtained, and how the findings translate into practice.

Prepared as a companion guide to the code repository *research/behavioral-finance/*
Written in English for readers without a quant-finance background; every concept is introduced with a plain-language example before it is used.

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Executive Summary

Financial theory has long assumed that markets are, to a first approximation, rational: prices reflect available information, and systematic mispricing should be rare and quickly corrected by sophisticated investors. This project tests that assumption from two directions at once, using real code and, wherever technically possible, real data rather than narrative alone.

Direction one (diagnosis): why do sophisticated, model-driven institutions sometimes fail catastrophically, precisely because their models assumed rational, well-behaved markets? Long-Term Capital Management (1998), the "Quant Quake" (2007), Amaranth Advisors (2006), and the 2008 financial crisis are examined as cases where a locally reasonable model broke because it could not see how fast correlation and panic move together once other market participants stop behaving independently.

Direction two (exploitation): can well-documented behavioral biases — anchoring, underreaction, overreaction — be turned into a measurable, tradable edge, the way real firms such as Fuller & Thaler Asset Management and LSV Asset Management have built businesses on doing? This project builds a signal library and backtest engine, and runs it against a real 21-year daily price history for 48 India-listed companies (the only market data source technically reachable from the environment this project was built in — see Part V for why that constraint mattered and how it was worked around).

The honest headline result from that empirical run: a blended "behavioral score" combining three signals looked like a roughly break-even strategy — and that blend was hiding two very different stories. One component signal (anchoring to the 52-week high) lost money persistently and dramatically; another (short-term reversal) at first looked like a real, positive, cost-adjusted edge. The project's central methodological lesson — never trust a blended score without checking what is inside it — came directly out of this finding, not from a textbook. (Chasing *why* the losing component lost money turned into the project's longest, most instructive thread — Part V walks through five rounds of increasingly rigorous testing. The losing signal's cause turned out simpler, and more humbling, than any of the behavioral explanations first considered — and the same rigor, once finally pointed at the "winning" signal too, retracted it and surfaced a different, genuine finding instead.)

The guide that follows explains every concept used (behavioral and quantitative) in plain language with real examples, walks through all five case studies in detail, narrates the project's milestones and the reasoning behind each decision, and translates the findings into an investment framework, a risk-management playbook, and a business idea.

Part I — Why This Project Exists

1.1 The theory markets are supposed to follow

The Efficient Market Hypothesis (EMH), the dominant framework in academic finance since the 1960s-70s, holds that asset prices reflect all available information at essentially every moment. Its practical implication is stark: if EMH holds strictly, no strategy should reliably beat a passive index over time, because any predictable pattern would immediately be traded away by rational, profit-seeking investors the moment it appeared.

EMH is not a strawman invented for this project to knock down — it is a genuinely useful default. Most of the time, for most large, liquid securities, prices really do incorporate public information quickly. The interesting question is not "is EMH ever true" (it often is, approximately) but "when and how does it fail, and can those failures be anticipated, measured, or profited from?"

1.2 Two stories that motivate the whole project

TWO EPISODES THAT FRAME THE PROJECT

In 1998, a hedge fund whose partners included two Nobel-laureate economists (Myron Scholes and Robert Merton, co-creators of the option-pricing theory taught in every finance program) lost about 90% of its capital in roughly six weeks, despite running trades its own models considered close to riskless. In January 2021, a struggling video-game retailer's stock rose by roughly an order of magnitude in a few weeks, not because of any change in the company's prospects, but because a large, loosely coordinated group of retail traders on a Reddit forum decided to buy it — and forced some of the most sophisticated short-selling hedge funds in the world to take large losses covering their bets. Both episodes are covered in full in Part IV.

These two stories point in opposite directions — one is about a sophisticated model failing, the other about an unsophisticated crowd winning — but they share a single underlying fact: humans, not equations, decide what a price is. A model can be mathematically elegant and still miss the fact that, under stress, people (whether panicking institutional traders or excited retail traders) start behaving in a correlated, non-independent way that no amount of historical calibration prepared it for.

1.3 The project's two-track design

Two concrete, falsifiable questions structure everything that follows:

- **Q1 (diagnosis).** Can the mechanism by which rational-market risk models fail under stress be shown mechanically, with real numbers, rather than just asserted narratively?
- **Q2 (exploitation).** Do specific, named behavioral biases produce a real, measurable, cost-adjusted return premium when turned into a systematic trading signal, over a long sample, on real market data?

DECISION & RATIONALE

The project could have been written as a pure literature review — summarizing what Kahneman, Tversky, Thaler, and others found, and describing LTCM and GameStop narratively. That would have been faster, but it would not have produced anything falsifiable: a narrative account can always be told to confirm whatever conclusion the writer wanted. The decision here was to require runnable code and, wherever technically possible, real executed numbers for every claim — even when (as happened repeatedly, see Part V) the environment made that harder than expected. A claim in this project is either backed by a number that was actually computed, or it is explicitly flagged as not yet validated. Nothing in between.

Part II — Core Behavioral Concepts, Explained Before They're Used

Every case study and every trading signal in this project rests on one of the concepts below. Each is introduced with an everyday example first, then the specific market mechanism it produces, so that later sections can use the term without re-explaining it.

2.1 Anchoring

Everyday example: if a car salesperson opens negotiation at \$30,000, most buyers end up settling closer to that number than if the same car had been opened at \$25,000 — even though the opening number carries no real information about the car's actual worth. The first number seen becomes a psychological reference point (an "anchor") that subsequent judgments are adjusted from, insufficiently.

Market mechanism: investors use a stock's 52-week high as exactly this kind of reference point. Even when good news arrives that should push the price well past the old high, many investors hesitate to buy through a level they've mentally anchored on as "expensive," so the price drifts up slowly instead of jumping — the mechanism behind the 52-week-high signal used in this project's Q2 signal library (Part III) and central to the George & Hwang (2004) academic finding it's based on.

2.2 Herding (Social Proof)

Everyday example: seeing a long line outside one restaurant and an empty one next door, most passersby join the line — not because they have independently evaluated either restaurant's food, but because a crowd is itself treated as evidence of quality. This is useful most of the time (crowds often do know something) and can be badly wrong when the crowd itself started for an unrelated reason.

Market mechanism: in a crisis, investors and institutions sell what everyone else is selling, and buy what's flowing toward safety (a "flight to quality"), regardless of the specific fundamental merits of what they're selling or buying. This is the central mechanism in the LTCM case (Part IV): LTCM's own trades were not herding, but the rest of the market's correlated flight to safety broke the correlation assumptions LTCM's model depended on.

2.3 Overconfidence and Illusion of Control

Everyday example: most drivers rate themselves as "above average" — a statistical impossibility for a majority to be true, and a well-replicated psychological finding. A string of good outcomes (safe drives, in this analogy) is easy to misread as evidence of skill rather than of favorable conditions that happened to hold.

Market mechanism: a trader with a strong track record under one market regime can mistake that track record for proof their model handles all regimes, and size positions accordingly. This is the central mechanism in the Amaranth Advisors case (Part IV): trader Brian Hunter's prior profitable natural-gas positions preceded a concentrated bet sized well beyond what the fund could survive being wrong about.

2.4 Underreaction and Overreaction

Everyday example: told a piece of surprising news, some people barely update their prior opinion at all (underreaction — "I'll believe it when I see more evidence"), while others swing to the opposite extreme immediately (overreaction — "this changes everything"). Both are documented, opposite failures of the same rational-updating standard (Bayesian updating: revising a belief by exactly as much as new evidence warrants, no more, no less).

Market mechanism, underreaction: after a company reports a surprising earnings beat, its stock price often keeps drifting upward for weeks afterward instead of jumping immediately to the new fair value — a well-documented anomaly called post-earnings-announcement drift (PEAD), because analysts and investors update their models too slowly. This underreaction is also the mechanism behind price *momentum* more generally: information diffuses slowly, so a trend already underway tends to continue.

Market mechanism, overreaction: a sharp, sudden price move (a panic sale or a euphoric rally) often partially reverses over the following weeks as the overreaction unwinds — the mechanism behind the short-term *reversal* signal used in this project, and, as it turned out, the single signal that actually worked in the real backtest (Part V).

2.5 Extrapolation Bias

Everyday example: a company that has grown fast for three years straight is often assumed, informally, to keep growing at a similar pace — even though few businesses sustain any given growth rate indefinitely. People extrapolate recent trends further into the future than the trends themselves justify.

Market mechanism: investors chase exciting recent growth stories ("glamour" stocks) and shun boring, currently out-of-favor companies, bidding growth stocks up beyond what their fundamentals justify and value stocks down below theirs. Lakonishok, Shleifer & Vishny's 1994 paper (the founding research behind LSV Asset Management, Part IV) argued this extrapolation bias, not extra risk, explains much of the historical outperformance of value stocks.

2.6 Attention and Availability Bias

Everyday example: people tend to overestimate the probability of dying in a plane crash relative to a car crash, because plane crashes are rare but vivid and heavily covered by media, so examples come to mind (are "available") more easily than the much more common, less-covered car crash.

Market mechanism: retail investors disproportionately buy stocks that have recently grabbed their attention — extreme returns, heavy news or social-media coverage — rather than researching broadly across the market first (Barber & Odean, 2008). This is the central mechanism behind the GameStop episode (Part IV): a stock became a coordinated retail target largely because it was already attention-grabbing, not because of a broad, independent fundamental re-evaluation by thousands of separate investors.

2.7 Crowding: Herding at Institutional Scale

Everyday example: if every restaurant critic in a city independently decides "fusion cuisine is the next big trend" and reviews accordingly, the city can end up with far more fusion restaurants than genuine demand supports — not because any critic copied another, but because they were all reasoning from similar assumptions and similar data.

Market mechanism: many quantitative hedge funds, working independently with different proprietary signals, can still end up holding very similar positions if their signals are built on the same underlying academic factors (value, momentum). Their portfolios look diversified from the inside but are secretly correlated with each other. This is the central mechanism behind the 2007 "Quant Quake" (Part IV): once one large fund was forced to sell, the selling pressure looked, to every other similarly-positioned fund's model, like a signal to sell the same names.

Part III — Quantitative Methodology, Explained

3.1 What is Value-at-Risk, and why does a "calm-regime" model miss the point?

Plain-language definition: Value-at-Risk (VaR) at the 99% confidence level answers the question "on a bad-but-not-extreme day (one of the worst 1% of days), roughly how much could this portfolio lose?" It is the financial -industry equivalent of an insurer asking "how much should we expect to pay out in a bad year, one we'd expect to see about once every hundred years?"

Why a Gaussian (bell-curve) model of VaR is dangerous: the easiest way to estimate VaR is to assume returns follow a normal (bell-curve) distribution, calibrated on however many years of historical data are available, and to assume the **correlations** between different positions — how much they tend to move together — stay roughly constant. Both assumptions are usually reasonable in ordinary conditions. Both usually fail together, at the worst possible time: during a genuine crisis, returns move by more than a bell curve predicts ("fat tails"), and positions that used to move mostly independently start moving together (correlations spike toward 1), because everyone is reacting to the same underlying fear at once.

ANALOGY: TWO HOUSES AND A WILDFIRE

Two houses on the same street, insured separately against fire. Ordinarily, one house catching fire tells an insurer almost nothing about whether the neighboring house will too — fires are close to independent events, so insuring both together looks safe. But a wildfire changes that: it can burn down both houses at once, because now the two risks share a single common cause. A risk model calibrated only on ordinary house fires will badly underestimate the insurer's exposure to a wildfire year, not because the model is bad at estimating ordinary fires, but because it never saw the correlation-raising event in its calibration data. This is exactly the mechanism this project's Part III.3 simulation quantifies for a leveraged trading book.

3.2 The Q1 risk simulation: setup

Rather than simply asserting that Gaussian VaR models understate tail risk, this project built a Monte Carlo simulation (*risk_simulation/fat_tails_vs_normal.py* in the code repository) that computes the underestimation directly. It models a leveraged, six-asset relative-value book (weights summing to roughly 9x gross exposure, loosely evocative of the kind of leveraged spread trades LTCM ran) under two return-generating processes:

- **The model the risk desk believes:** a Gaussian (bell-curve) distribution, with a modest, constant 0.25 pairwise correlation between assets, calibrated on "calm regime" data — exactly the kind of risk model a desk would build from a few years of ordinary historical returns.
- **The process that actually generates the data:** the same calm regime most days, but with a small (1%) daily probability of switching into a "stress regime" where volatility roughly triples and pairwise correlation jumps to 0.92 — combined with fat-tailed (Student-t) shocks in both regimes for realistic extreme-day behavior.

DECISION & RATIONALE

This project's sandbox environment had no general internet access when this simulation was built (see Part V for the full story), which ruled out downloading real historical crisis data to measure this effect directly. Rather than skip the question or fake a result, the decision was to build a Monte Carlo simulation that needs no external data at all — every number it reports is a genuine, reproducible output of code that actually ran, even though the underlying scenario is stylized rather than a literal reconstruction of any single historical event. The parameters (0.25 to 0.92 correlation, 3x volatility) were chosen to be qualitatively consistent with what is publicly known about the LTCM episode's shape, not fit to LTCM's actual (never fully disclosed) book — the report is explicit that the resulting multiple should be read as "this is the order of magnitude of the effect," not a precise historical reconstruction.

3.3 The Q1 results (real, executed numbers)

Tail depth	Gaussian (calm-regime) VaR	True (simulated) VaR	True Expected Shortfall (CVaR)
99.0%	7.67% of book	8.15% of book (1.06x)	10.37% of book (1.35x)
99.9%	10.19% of book	13.26% of book (1.30x)	16.83% of book (1.65x)

Worst single day across 500,000 simulated trading days: a loss of 58.9% of the book — a scale of move a Gaussian, calm-regime model treats as essentially impossible, but which falls naturally out of a process that occasionally lets volatility and correlation rise together. The gap between the Gaussian model and reality is modest right at the 99% line (that quantile happens to sit near the boundary between regimes in this parameterization) but compounds quickly deeper into the tail — which is itself an important lesson: a risk model can look adequate at the confidence level a regulator or risk committee checks routinely, and still be dangerously wrong exactly where it matters most, a few notches further into the tail.

3.4 Turning behavioral biases into tradable formulas

Part II described three biases in plain language. Here is how each becomes a number that can be computed for every stock, every day, and used to rank a universe of companies:

Signal	Formula (plain language)	Bias it targets
12-1 month momentum	Cumulative return over the trailing ~12 months, excluding the most recent month (to avoid contaminating the trend)	Underreaction
52-week-high proximity	Current price divided by the highest price reached in the trailing 52 weeks (1.0 = at the high)	Anchoring
Short-term reversal	Negative of the trailing ~1-month return (a recent loser scores positively, betting on a bounce)	Overreaction

A **composite score** blends all three by converting each to a cross-sectional z-score (how many standard deviations a stock is above or below the average stock, on that signal, on that day) and averaging — the intended purpose being a single number that combines all three biases into one rank. Part V explains why this turned out to be a more dangerous simplification than it first appears.

3.5 How the backtest actually works

The methodology follows the standard approach used in the academic momentum and reversal literature, so results are comparable to published findings rather than being an artifact of a bespoke design:

- Once a month, every stock in the universe is ranked by its current signal score.
- The strategy goes equally-weighted *long* the top group (the best-ranked names) and equally-weighted *short* the bottom group (the worst-ranked names) — a "long-short" portfolio, so its return reflects the spread between the best and worst names, not the market's overall direction.
- The portfolio is held for one month, then re-ranked and rebalanced.

Three metrics summarize the result:

- **Sharpe ratio:** average return divided by the volatility (bumpiness) of that return, annualized. A Sharpe of 1.0 is generally considered good for a systematic strategy; a Sharpe near 0 means the strategy earned close to nothing per unit of risk taken; a negative Sharpe means it lost money relative to its own risk.
- **Maximum drawdown:** the largest peak-to-trough decline the strategy experienced at any point in the backtest — a measure of the worst single stretch an investor in the strategy would have had to sit through.
- **Turnover and transaction costs:** every rebalance that changes which names are held costs money in practice (bid-ask spread, market impact). The backtest engine tracks how much of the portfolio changes each month (turnover) and deducts a simple cost (basis points per unit of turnover) to report both a "gross" return (before costs) and a "net" return (after costs) — because a strategy that looks good gross and disappears net of realistic costs is a very common way quantitative research misleads itself.

Part IV — Five Case Studies, in Depth

Each case below follows the same structure: what happened, with a timeline; which behavioral mechanism from Part II explains it, in plain language; and why the case matters for this project's central question.

4.1 Long-Term Capital Management (1998)

What happened

Long-Term Capital Management was a hedge fund founded in 1994 by John Meriwether, with partners that included Myron Scholes and Robert Merton, who shared the 1997 Nobel Memorial Prize in Economic Sciences for the Black-Scholes-Merton option-pricing framework taught in every finance curriculum. The fund ran highly leveraged convergence/relative-value arbitrage: betting that prices of closely related securities would converge toward their historically "fair" relationship. Individual trades had small expected edges, so the fund used enormous leverage — over 25:1 on its balance sheet, with derivative notional exposure exceeding \$1 trillion — to turn small, statistically reliable-looking edges into a large return.

From 1994-1997 the fund returned roughly 20-40% annually, a track record that built exactly the kind of confidence Part II.3 (overconfidence) describes. Russia's default and rouble devaluation on August 17, 1998 triggered a global "flight to quality": investors worldwide fled toward the safest, most liquid assets (US Treasuries) and away from everything else, regardless of the specific fundamental link between the assets they were selling. Spreads LTCM's model treated as largely independent instead moved together, in the same direction, and kept widening instead of converging. The fund lost about \$1.9 billion in August alone; by September 21, capital had fallen from roughly \$4.7 billion at the start of the year to about \$600 million, against essentially unchanged notional exposure. On September 23, 1998, with the fund days from a disorderly default that regulators feared could force a fire-sale across Wall Street, the Federal Reserve Bank of New York organized (but did not fund) a \$3.625 billion recapitalization by 14 banks in exchange for 90% of the fund's equity.

The behavioral mechanism, explained

This is a herding case (Part II.2), but with an important twist: the herding was not LTCM's own. LTCM's individual trades were, in isolation, defensible. What broke the model was that every other market participant started behaving in a correlated way at once — a collective flight to safety that a "rational agents, independent decisions" model does not naturally produce. The lesson generalizes: a model of rational markets is only as good as its assumption that other participants keep behaving independently under stress, and that is precisely the assumption that breaks in a genuine crisis — the exact mechanism quantified in Part III's Q1 simulation.

Why it matters here

LTCM is the founding case for this entire project: a fund staffed by the people who literally wrote the mathematics of modern option pricing still lost 90% of its capital in six weeks, not because the mathematics was wrong, but because the mathematics assumed away exactly the kind of correlated human behavior that shows up under real stress.

4.2 The "Quant Quake" (August 2007) and Amaranth Advisors (2006)

What happened — Quant Quake

Between roughly August 6-9, 2007, a large number of market-neutral quantitative equity funds — using different individual signals, run by different firms, with no apparent common trigger in company fundamentals — suffered simultaneous, sharply correlated losses over a few trading days, followed by a partial rebound once the selling pressure eased. Several well-known quant strategies lost on the order of 5-10%+ in that single week — an extreme move for supposedly market-neutral, low-volatility books. The standard academic reconstruction (Khandani & Lo, 2007/2011) attributes this to **crowding**: many quant managers had, independently, arrived at similar factor tilts (value, momentum), so their books were far more correlated with each other than any of them individually realized. When one or more large funds were forced to deleverage a quant book quickly (for reasons believed unrelated to those specific positions), the forced selling looked, to every other similarly-built model, like a signal to sell the same names.

What happened — Amaranth

Amaranth Advisors was a large multi-strategy hedge fund that lost approximately \$6.5 billion in September 2006, almost entirely from concentrated, highly leveraged natural-gas futures and options positions run by a single trader, Brian Hunter. Hunter's core position was a calendar spread — long winter-month natural-gas futures against short summer-month contracts, betting the spread would widen — sized large enough, relative to actual market open interest, that Amaranth's own trading could move the market it was betting on. A mild start to the 2006 hurricane season pushed the spread the wrong way; over roughly a week in mid-September 2006 the fund lost more than half its capital.

The behavioral mechanisms, explained

Quant Quake is a crowding case (Part II.7): herding among the most sophisticated participants in the market, the group a naive efficient-markets story would expect to be immune to it. Amaranth is an overconfidence case (Part II.3): Hunter's prior profitable natural-gas trades made a concentrated, survivable-only-if-right position look more justified than the fund's risk limits should have allowed.

Why they matter here

Together, these two cases show that neither sophistication (Quant Quake) nor a genuinely skilled track record (Amaranth) protects against the specific behavioral failure modes this project studies — they just change which failure mode shows up.

4.3 The 2008 Financial Crisis and the Gaussian Copula

What happened

The 2007-2008 crisis is usually told as a subprime mortgage story, but the mechanism that turned a real but containable housing problem into a systemic banking crisis mirrors this project's Q1 simulation almost exactly. Banks packaged mortgage pools into collateralized debt obligations (CDOs), tranching by seniority. A senior tranche's safety depends critically on how correlated the underlying mortgage defaults are: independent defaults make a senior tranche very safe; correlated defaults (a national downturn hitting every region at once) make that protection evaporate. The industry-standard tool for pricing this correlation risk was the Gaussian copula (David X. Li, 2000), which reduced the hard problem of modeling thousands of correlated defaults to a single correlation parameter, estimated from a history drawn almost entirely from a multi-decade housing boom during which national home prices had never fallen.

When US home prices began falling nationally in 2006-2007 — a regime the calibration window had essentially no examples of — mortgage defaults stopped being independent and started moving together, for the same underlying reason, in every region at once. Senior tranches priced as extremely safe took losses far larger and faster than the model implied, and because these instruments were held throughout the banking system, the losses hit systemically important institutions almost simultaneously (Bear Stearns, March 2008; Lehman Brothers' bankruptcy, September 15, 2008; AIG's roughly \$182 billion federal rescue, driven largely by credit default swaps written against these same instruments).

The behavioral mechanism, explained

This is structurally the identical mistake as a calm-regime Gaussian VaR model (Part III.1's wildfire analogy): a correlation parameter estimated from a period that never included the correlation-raising regime, and therefore blind to how fast that correlation could rise once it arrived. Two behavioral/institutional layers sit on top of the pure model failure: **herding into a shared industry model** (using it was a competitive necessity even for desks that suspected it understated risk — a rational individual response to career/franchise risk producing an irrational collective outcome), and a **correlated trust collapse** once losses started (nobody could tell who was exposed to how much bad paper, so short-term lenders pulled back from everyone at once — the same flight-to-quality mechanism as LTCM, at systemic scale).

Why it matters here

This is the largest-scale version, in modern financial history, of the pattern this project's Q1 simulation makes explicit in miniature: a model that is locally reasonable, calibrated on real data, and wrong specifically about how correlation behaves in the rare regime that does the damage.

4.4 GameStop and the "Meme Stock" Squeeze (January 2021)

What happened

Through 2020, several large hedge funds, most prominently Melvin Capital, held large short positions in GameStop (GME), a struggling brick-and-mortar video-game retailer, on a conventional view that the business was in structural decline. Starting in mid-to-late January 2021, a large, coordinated wave of retail buying — organized largely on the Reddit forum r/wallstreetbets — pushed the share price up by roughly an order of magnitude in a few weeks. Two mechanisms compounded pure buying pressure: a **short squeeze** (as the price rose, short sellers faced margin calls and were forced to buy shares to close losing positions, pushing the price up further) and a **gamma squeeze** (heavy buying of short-dated call options forced the market-makers who sold those calls to buy the underlying stock to stay hedged, an amplifying loop running through market structure rather than anyone's fundamental view). Melvin Capital's short position lost the fund a very large share of its capital in January 2021 and received a \$2.75 billion capital injection from Citadel and Point72 on January 25, 2021. Robinhood and other brokerages restricted buying of GameStop and similar stocks on January 28, 2021, citing clearinghouse deposit requirements — a decision that triggered US congressional hearings in February 2021.

The behavioral mechanisms, explained

This case runs the *opposite* direction from the others: retail herding (Part II.2, at internet speed and scale) overwhelming institutional positioning, amplified by attention-driven trading (Part II.6 — GameStop became a target largely because it was already attention-grabbing, consistent with Barber & Odean's research on individual investors disproportionately buying stocks that recently grabbed their attention) and by the gamified design of mobile-first trading apps.

Why it matters here

GameStop is included specifically to prevent this project's thesis from collapsing into "institutions are rational, retail is irrational" — a genuinely coordinated crowd can overwhelm sophisticated positioning just as forcefully as institutional herding overwhelmed LTCM's models, just pointed in the opposite direction.

4.5 The Other Side: Funds Built to Exploit These Biases

Every case above is a story about a bias causing a loss. These three firms are the counter-evidence: specific, named biases treated as a persistent, exploitable source of return.

Fuller & Thaler Asset Management

Founded in 1993 by Richard Thaler — awarded the 2017 Nobel Memorial Prize in Economic Sciences for work on how limited rationality affects economic decisions — together with Russell Fuller. Their signature mechanism is closely related to Part II.4's underreaction concept: post-earnings-announcement drift, the well-documented finding that stock prices keep drifting in the direction of an earnings surprise for weeks after the announcement, because analysts anchor on their pre-announcement estimates and update too slowly. Their flagship retail vehicle has long traded as "Undiscovered Managers Behavioral Value" (ticker UBVLX).

LSV Asset Management

Founded in 1994 by academic economists Josef Lakonishok, Andrei Shleifer, and Robert Vishny, whose own research (Lakonishok, Shleifer & Vishny, *Journal of Finance* 49(5), 1994, pp. 1541-1578) argued that much of value stocks' historical outperformance over "glamour" growth stocks is explained by investors extrapolating recent growth too far into the future (Part II.5), not by value stocks simply being riskier.

AQR Capital Management

Founded in 1998 by Cliff Asness, whose University of Chicago PhD dissertation (under advisors including Eugene Fama, himself closely associated with the efficient markets hypothesis) was on momentum in stock returns. AQR treats behavioral anomalies as one input among several (alongside risk-based explanations), run with heavy quantitative discipline — a useful middle case between Fuller & Thaler's purely behavioral framing and a pure risk-based factor story.

The working assumption behind all three — and behind this project's signal library in Part III — is that specific, named biases produce persistent, not merely historical, mispricing. Part V.2's empirical result tests that assumption directly, with a real, non-cherry-picked outcome.

Part V — Project Journal: Milestones, Decisions, Results

This part narrates the project as it actually happened, in order, including the constraints that forced decisions and the results that changed the project's direction. Every decision below is stated together with the reasoning behind it, per this guide's brief: nothing is presented as self-evidently correct without the "why."

5.1 Scoping the project

The project started from a broad brief: research the role of human irrationality in markets, with concrete applications to finance and business. Before writing anything, the scope was narrowed by asking four questions: what output format was wanted, which behavioral angle to anchor on, what methodology/evidence to use, and what concrete impact the research should produce.

DECISION & RATIONALE

The answers chosen were: a **quantitative research project with runnable code** (not a pure literature review); anchored on **both** failed "rational market" bets (LTCM-style) **and** behavioral funds that exploit bias; using **public market data plus case-study comparison**; aimed at producing an **investment framework, risk-management lessons, and a business idea**. This combination was chosen specifically because it forces the project to be falsifiable in both directions at once (Part I.3) rather than only telling the flattering half of the story (either "markets are irrational, here's how to profit" or "here's why smart people fail," without testing either claim against real numbers).

5.2 Milestone 1 — Building the pipeline under a hard constraint

The first concrete obstacle appeared immediately: the development environment's outbound network access is restricted by an egress proxy to a specific allowlist. Direct testing (curl requests to each host, and reading the proxy's own status endpoint, which logs connection failures) confirmed that Yahoo Finance, Stooq, SEC EDGAR, and FRED — every conventional free market-data source — were all blocked with a 403 policy denial.

DECISION & RATIONALE

The tempting shortcut here would have been to generate synthetic "market-looking" data and present backtest results computed on it as if they were real findings — exactly the kind of model-dressed-as-reality mistake this project's whole thesis warns against. The decision instead was to build the complete, real pipeline (data loader, signal library, backtest engine) as production-quality code, but to sharply separate two categories of claim: results that were **actually computed by running code in this environment** (which get stated as fact) versus code that is **complete and believed correct but not yet exercised on real data** (which gets stated as an explicit, flagged limitation, never as a finding). The one part of the research that needed no external data at all — the Q1 tail-risk simulation (Part III.2-3.3) — was built first specifically because it could produce a genuine, real result despite the network constraint.

Milestone 1 shipped: the research design document, the full signal and backtest codebase (validated only against synthetic test fixtures at this point, with that limitation stated explicitly), the Q1 risk simulation with its real executed numbers, three initial case studies (LTCM, Quant Quake / Amaranth, the behavioral funds), and a first version of the practical framework document.

PROCESS DECISION

At the end of milestone 1, a new working practice was established at the user's request: after every milestone, provide a recap of the reasoning behind what was built, followed by explicit questions checking whether the project's focus was still correctly calibrated to what was actually wanted. This is why milestones 2 and 3 below each open with what the previous recap's questions surfaced.

5.3 Milestone 2 — Real data, and a result that changed the project

The milestone-1 recap asked, among other things, how to handle the no-internet constraint going forward. The chosen answer was to actively search for a reachable data source rather than accept the limitation as final.

DECISION & RATIONALE

Rather than assume every external host was blocked, a range of unrelated hosts was tested directly — not just financial-data providers but also generic ones (a plain package index, a generic search engine). This showed the block was not "no internet at all" but a specific allowlist: *api.github.com* and *raw.githubusercontent.com* were reachable while general financial-data hosts and even a plain general-purpose website were not. That distinction mattered: it meant GitHub-hosted datasets were a real option, not a dead end.

A web search for GitHub-hosted multi-ticker daily price datasets surfaced a real candidate: a community-uploaded CSV of 63 raw ticker symbols covering NSE (India) large/mid-cap companies from 2000 to 2021. Before using it, the data was checked for basic quality (no duplicate date/symbol rows, no zero or negative prices) and then for a subtler issue: several apparent "different companies" turned out,

on inspection of their trading date ranges, to be the *same* company under a former ticker symbol.

DECISION & RATIONALE

The evidence for this was date-range contiguity: symbol A's last trading date and symbol B's first trading date lined up with no overlap and no gap, for 14 separate symbol pairs (or longer chains) — the exact signature of an NSE ticker rename. Each chain was then cross-checked against publicly known Indian corporate history (for example, "TELCO" was Tata Motors' ticker before a rename; "SESAGOA" became "SSLT" then "VEDL" as Sesa Goa became Sesa Sterlite became Vedanta Ltd) before merging. One pair, HDFC and HDFCBANK, was deliberately *not* merged, because the two were genuinely separate listed companies for the entire sample period (they only merged in 2023, after the data ends). This is the kind of judgment call that is easy to get wrong silently in a backtest — merging two unrelated companies, or failing to merge one company's own history — so it was done from evidence, and locked in afterward with a dedicated automated test (visible in the code repository's `tests/test_loaders.py`) rather than left as a one-off manual step.

With 48 clean, continuous company series covering 21 years, the full signal-and-backtest pipeline was run for the first time on real market data.

Signal	Split	Gross annual return	Gross Sharpe	Net Sharpe	Max drawdown
Composite (all 3)	deciles	-3.59%	0.01	-0.03	-74.3%
Composite (all 3)	quintiles	-1.33%	0.05	0.00	-63.2%
12-1 momentum only	quintiles	-1.73%	0.04	0.01	-60.8%
52-week-high only	quintiles	-13.71%	-0.51	-0.54	-97.1%
Short-term reversal only	quintiles	+4.46%	0.30	0.22	-64.1%

The composite score alone would have supported a bland, roughly-correct-sounding conclusion: "no real edge here." Decomposing it into its three parts told a much more informative and, in one case, misleading story: the 52-week-high (anchoring) signal was actively destructive, losing money persistently and suffering a 97% drawdown, while the short-term reversal (overreaction) signal *looked* like a genuine, positive, cost-adjusted edge. Averaging them together hid both facts — and Part V.9 (Milestone 8) later found the reversal side of that story didn't survive the same scrutiny the losing side got.

A follow-up check tested whether the anchoring signal's damage was simply an artifact of the 2008 and 2020 crashes (both periods where a "buy recent winners" strategy is known from the academic literature to be exposed to sharp reversals — "momentum crash risk"). Excluding both crash windows entirely, the strategy's Sharpe ratio outside them was still -0.46, and its cumulative loss *excluding* the crashes (-90%) was actually larger than its loss *during* them (-53%) — ruling out "it only lost money in the two crashes" as the explanation, and leaving the true cause an open question flagged honestly rather than resolved with a plausible-sounding guess. (Part V.4 below picks this exact question back up.)

DECISION & RATIONALE

The milestone-2 recap surfaced this same lesson back to the practice guiding the project itself: a mediocre-looking blended result should always be decomposed before being trusted, a rule now stated explicitly in the practical framework (Part VI.1). The recap also asked whether tying the eventual business idea to this repository's own existing software platform (a private-equity deal-screening tool the code happened to live alongside) was the right framing. The answer was no — the business idea was rewritten from scratch as a standalone analytics service (Part VI.3), decoupled from any specific existing platform, per that explicit feedback.

5.4 Milestone 3 — Replication, and a rejected hypothesis

The milestone-2 recap's questions produced three directions: search for a second, independent market's data to check whether the NSE findings replicate elsewhere; investigate *why* the 52-week-high signal lost money, specifically testing whether it was confounded with a value/growth effect; and continue expanding the codebase rather than wrapping up. Producing this guide itself was folded into the same milestone, at the user's explicit request to run it in parallel with the quantitative work rather than after it — which is why an earlier draft of this section reported milestone 3 as in-progress; both investigations below have since completed.

Finding a second market

A GitHub-hosted mirror of the well-known Kaggle "Huge Stock Market Dataset" (one CSV per US-listed ticker) was located and verified: 30 US large-cap names, the same tickers as the project's original default universe, each starting at its own listing date and running through 2017-11-10 (the dataset's last snapshot; this is a historical replication check, not a live feed).

Signal	Split	Gross annual return	Gross Sharpe	Net Sharpe	Max drawdown
Composite (all 3)	quintiles	+2.99%	0.25	0.21	-79.6%
12-1 momentum only	quintiles	+5.72%	0.34	0.32	-88.4%
52-week-high only	quintiles	-13.98%	-0.29	-0.32	-99.9%
Short-term reversal only	quintiles	-0.68%	0.13	0.07	-83.0%

The result was not a clean confirmation of the NSE finding — it was more informative than that. **Momentum and reversal flipped which one looked real:** reversal was NSE's edge and the US market's near-zero result; momentum was NSE's near-zero result and the US market's real, positive, cost-adjusted edge (Sharpe 0.32 net). Only one thing held up unchanged across both very different markets and eras: **the 52-week-high signal lost money in both.** (Part V.9, Milestone 8, later put both sides of this flip through a beta-adjusted significance test: the US momentum side held up and then some; the NSE reversal side did not.)

DECISION & RATIONALE

A single-market backtest result is provisional by default. Had this project stopped after milestone 2 (NSE only), it would have reported "momentum is dead here, reversal is the real edge" — the opposite of what the US data alone would have suggested. Neither market alone is the right answer; the honest finding is that anomaly presence is universe-dependent for momentum and reversal, while the 52-week-high signal's damage is not.

Testing the value/growth confound — and rejecting it

The leading candidate explanation for the 52-week-high signal's losses, carried over from milestone 2, was that India's two-decade secular bull market could make "short the names far from their 52-week high" equivalent to "short cheap, beaten-down names" that then mean-revert upward — a value effect masquerading as an anchoring effect. This was tested directly: a simple price-only value proxy (current price divided by its own trailing ~3-year average) was built, its cross-sectional correlation with the raw 52-week-high score was measured, and the 52-week-high signal was then **orthogonalized** against it — a date-by-date cross-sectional regression that mathematically removes the shared value component, leaving only the part of the signal that isn't explained by it.

Market	Correlation (signal, value proxy)	Raw signal net Sharpe	Orthogonalized net Sharpe
NSE (India)	0.63	-0.54	-0.73
US (Kaggle mirror)	0.44	-0.32	-0.49

The two signals genuinely are correlated (0.44-0.63) — a stock near its 52-week high does tend to look "expensive" on this proxy. But removing that shared component made performance **worse**, not better, in both markets. If the value/growth confound had been the real explanation, the "purified" signal should have looked neutral or better once the confounding value component was stripped out; instead it got more negative. **The value/growth confound hypothesis is rejected.**

WHAT THIS DOES AND DOESN'T PROVE

This is a negative result, and it was reported as one rather than quietly dropped or reframed as a success. Combined with the milestone-2 finding that the loss isn't concentrated in the 2008/2020 crash windows either, two of the three candidate explanations for the 52-week-high result are now ruled out by direct test, not just by argument. The remaining candidate — generic momentum-crash risk, a documented property of "long recent winners" strategies in general — is the best-supported explanation left standing, but it has not itself been directly tested (that would mean checking whether the losses cluster specifically in high-volatility periods). The project states this as narrowed, not solved.

5.5 Milestone 4 — Testing momentum-crash risk directly

READ THIS FIRST

Update (Section 5.6, Milestone 5): the formal statistical test does NOT confirm the regime-conditioning mechanism this section describes. This section is kept as written at the time — the pattern-matching genuinely looked this compelling — but Section 5.6 walks it back with real numbers. Read that section before treating anything below as confirmed.

The one remaining candidate explanation from milestone 3 — generic momentum-crash risk (Daniel & Moskowitz, "Momentum Crashes," 2016; see Part II.4 and Part IV.1) — was tested directly at the user's request. The theory's specific prediction: a "long recent winners, short recent losers" strategy is exposed to severe losses when realized market volatility is elevated and the market has recently been in a downturn, because the *short* leg (recent losers, typically higher-beta) can rebound sharply in a recovery — and the damage should concentrate specifically in that short leg, not be spread evenly across both sides of the trade.

Two things were built to test this properly rather than just plausibly: two regime indicators from the same price data (a realized-volatility tercile and a trailing-12-month bull/bear flag, both lagged one day to avoid look-ahead), and an extension to the backtest engine itself so it reports the long leg's and short leg's profit-and-loss contributions *separately*, not just their combined total — needed to check the mechanism, not only the timing.

Evidence	NSE (India)	US (Kaggle mirror)
Sharpe vs. realized-vol tercile (Low/Mid/High)	-0.22 / -0.41 / -0.78	-0.11 / -0.27 / -0.43
Sharpe, bull vs. bear trailing-12m state	-0.27 vs. -1.06	-0.15 vs. -1.10
Worst bucket: bear + high-vol	-1.07 Sharpe, -38% ann.	-1.26 Sharpe, -61% ann.
Return skewness, combined long-short	-0.66	-1.32

All four signatures point the same way, in both markets. The leg decomposition confirms the mechanism, not just correlated timing: the **long leg (buying stocks near their 52-week high) is a genuinely positive, standalone strategy** in both markets (Sharpe 0.45-1.03 in every regime tested on NSE; 0.65-1.20 outside bear markets in the US) — the anchoring/underreaction thesis actually works on the long side. **The short leg is the entire problem:** negative in every single regime bucket in both markets, and its losses are specifically what swells in the bear-plus-high-volatility bucket (NSE: -19% to -49% annualized from calmest to worst regime; US: -13% to -62%).

HOW HARD TO LEAN ON THIS, AS UNDERSTOOD AT THE TIME

This is pattern-matching against a well-documented mechanism using real data across two independent markets, not a formal significance test — no statistical t-tests were run on the regime differences, and the volatility-tercile cutoffs were computed over each dataset's full history rather than a rolling window a live system would need. With that caveat stated plainly, at this point in the project four independent, mutually consistent signatures looked like about as strong a case as this kind of historical analysis could make short of formal statistics — which is exactly what Section 5.6 adds, and exactly what changes the conclusion.

5.6 Milestone 5 — Formally testing momentum-crash risk

The user explicitly asked for the pattern-matching in Section 5.5 to become "a statistical test with proven reliability" before opening a pull request. Two upgrades were made (*investigations/momentum_crash_significance.py*): the volatility tercile threshold now uses an **expanding window** (each day's "high vol" label uses only data through the previous day, fixing the one look-ahead caveat Section 5.5 flagged), and strategy returns are regressed on the regime indicators using **Newey-West (HAC) standard errors** — the standard correction for serial correlation from monthly-rebalanced, overlapping holding periods, the same style of correction the original Daniel & Moskowitz paper itself uses. Tests were run at both daily frequency (HAC lag 21) and monthly frequency (one return per rebalance, HAC lag 6, matching the paper's own testing frequency).

Return series	Daily: high_vol x bear	Monthly: bear	Monthly: vol_rank
NSE short leg (theory's predicted locus)	p=0.57	p=0.67	p=0.09 (wrong sign)
NSE long leg	p=0.94	p=0.0007	p=0.0046
US short leg (theory's predicted locus)	p=0.65	p=0.34	p=0.61
US long leg	p=0.15	p<0.0001	p=0.0139

The specific regime-conditioning mechanism does not survive formal testing. The short leg — the one momentum-crash theory specifically predicts should blow up in high-volatility, bear-market regimes — shows no statistically significant regime-conditioning in either market, at either frequency, in any specification tested. NSE's short leg even has a wrong-signed coefficient at the 10% level (losses *shrinking*, not growing, as volatility rises) — evidence against the hypothesis in that specific cut, not for it. The regime effects that *are* highly significant ($p < 0.01$, every specification) show up in the long leg instead — a far more mundane explanation (a long-biased position carries positive market-beta exposure and underperforms generally in bear markets) than the specific short-squeeze mechanism the hypothesis describes.

WHAT SURVIVES AND WHAT DOESN'T, AS UNDERSTOOD AT THIS POINT

What remains statistically ironclad, every specification, both markets, both frequencies: the long leg's average return is significantly positive and the short leg's is significantly negative — the decomposition itself, not the proposed explanation for it. With roughly 24 regime-effect coefficients tested across both frequencies and markets, seeing 2-4 marginally significant results at the 5-10% level is within what pure chance produces under a true null — not meaningfully more than a false-positive rate. As understood at this point, the long-only recommendation still stood — but the project cannot claim to know *why* the short leg fails. That remained genuinely open, and Section 5.7 is where it actually gets resolved, one more check further.

5.7 Milestone 6 — Is it just beta?

Milestone 5 left one basic check untried: does the long leg and the short leg simply have different, uncontrolled exposure to the market itself? An equal-weighted decile long-short book makes no attempt to match the two baskets' market sensitivity. If "losers" (far from the 52-week high) happen to be higher-beta stocks than "winners" (near the high), the strategy carries unintended net market exposure — and in a market with a strongly positive average return over the sample (both NSE and the US mirror returned roughly 19-21% annualized over their windows), that alone would produce exactly the pattern seen, with no anchoring or crash story required.

A standard CAPM-style regression was run (*investigations/short_leg_beta.py*): $\text{leg return} = \alpha + \beta \times \text{market return}$, using the same equal-weighted market proxy and the same Newey-West correction as Section 5.6, at daily and monthly frequency, for the long leg, the short leg, and the combined book.

	NSE daily	NSE monthly	US daily	US monthly
Long leg beta (p)	+0.84 (p<.0001)	+0.80 (p<.0001)	+0.82 (p<.0001)	+0.75 (p<.0001)
Short leg beta (p)	-1.16 (p<.0001)	-1.19 (p<.0001)	-1.39 (p<.0001)	-1.40 (p<.0001)
Combined net beta (p)	-0.32 (p<.0001)	-0.40 (p=.0007)	-0.57 (p<.0001)	-0.70 (p<.0001)
Long leg alpha (p)	p=0.60	p=0.61	p=0.76	p=0.35
Short leg alpha (p)	p=0.15	p=0.65	p=0.72	p=0.95
Combined alpha (p)	p=0.25	p=0.42	p=0.69	p=0.41

This is the answer. All 12 beta coefficients (both legs plus the combined book, both markets, both frequencies) are highly significant (p<0.001, all but one below p=0.0001). All 12 alphas are statistically indistinguishable from zero (p from 0.15 to 0.95). The "losers" basket consistently carries a larger-magnitude beta than the "winners" basket — consistent with the momentum literature's own description of loser-leg stocks as higher-beta — but the consequence here is purely mechanical: **the combined book is not market-neutral. It carries a significant, unintended net short-beta position (-0.32 to -0.70), and being structurally short a market that returned ~20% a year is a completely sufficient explanation for the losses documented in every milestone above — no anchoring bias,**

no crash risk, no value confound required.

WHAT THIS CHANGES

This changes the practical recommendation from Sections 5.5-5.6, not just adds a footnote. "Trade the signal long-only" rested on the long leg's raw average return being significantly positive. That return is real, but this test shows it's consistent with simply holding an 0.75-0.84-beta long position in a rising market, with **no significant standalone alpha** once that exposure is controlled for. There is currently no evidence, anywhere in this project, of genuine stock-selection skill in the 52-week-high signal, in either direction, once market beta is properly accounted for. The honest recommendation: this signal's long leg is a reasonable lower-beta way to stay invested, not a demonstrated behavioral edge. Testing for real, beta-independent skill would require explicitly beta-neutralizing both legs and re-running this same regression on the hedged residual — done next, in Section 5.8.

THE LESSON, AGAIN

The real methodological lesson: three milestones (2, 3, 5) chased increasingly sophisticated behavioral and statistical explanations — crash windows, a value/growth confound, formal regime-conditioning significance tests — before checking the single most basic hygiene check for any long-short equity backtest: whether the two legs are beta-matched. They weren't, and that omission alone explains everything the more exotic hypotheses were built to explain. Order hypothesis tests from cheapest and most mechanical to most exotic, not the reverse.

5.8 Milestone 7 — Does real alpha survive an actual beta hedge?

Milestone 6 used one full-sample regression coefficient to estimate beta and asked whether the leftover average return was significantly different from zero. That is a legitimate test, but it assumes constant market exposure across 20+ years, which real betas don't have. This milestone builds and tests the real thing (*investigations/beta_hedged_backtest.py*): at every monthly rebalance, beta is re-estimated from only the **trailing 252 trading days** (roughly one year) of the combined book's own history — strictly out-of-sample, never using data from the period being hedged — and that beta is used to hedge the *next* month's daily returns. This mirrors how a real fund would operationally hedge: periodic re-estimation applied forward, no look-ahead.

	NSE (India)	US (Kaggle mirror)
Days covered (needs 252d history)	5,142	11,417
Rolling hedge beta: mean (std)	-0.26 (0.31)	-0.44 (0.50)
Correlation of hedged returns w/ market	-0.07	-0.02
Unhedged ann. return / vol / Sharpe	-14.3% / 23.9% / -0.53	-14.7% / 33.5% / -0.30
Hedged ann. return / vol / Sharpe	-7.1% / 22.7% / -0.21	-7.5% / 31.3% / -0.09

	NSE (India)	US (Kaggle mirror)
Hedged daily return significant?	p=0.35	p=0.45
Hedged monthly return significant?	p=0.42	p=0.17

Independent confirmation of Milestone 6, by a stronger method. The rolling hedge substantially cuts the strategy's correlation with the market (from strongly negative down to -0.02 to -0.07) and **roughly halves the annualized loss** in both markets — consistent with a large share of the original loss being mechanical beta exposure, not something specific to the "losers" basket. But even after this real, out-of-sample hedge, **the residual return is not statistically distinguishable from zero in either market, at either frequency** (p between 0.17 and 0.45). No significant alpha, positive or negative, survives once market exposure is genuinely — not just statistically — removed.

TWO HONEST CAVEATS

Two honest caveats. First, the hedge is imperfect: residual correlation isn't exactly zero, and the rolling beta itself is quite unstable over time (its standard deviation is comparable to or larger than its mean in both markets) — a real hedging program would need frequent rebalancing and would incur transaction costs this script doesn't model. Second, a real, textbook bug was caught and fixed before any of these numbers were reported: an early version estimated beta as `np.cov(...) / np.var(...)`, but NumPy's `cov` defaults to a different degrees-of-freedom convention (`ddof=1`) than its `var` (`ddof=0`) — silently inflating every beta estimate by roughly $n/(n-1)$ (about 0.4% here, immaterial to the conclusion, but a real bug regardless). Fixed by using an OLS slope instead, and locked in with a synthetic-fixture test asserting exact beta recovery on a zero-noise series.

5.9 Milestone 8 — Does the reversal edge survive a beta check too?

Every beta check so far (Sections 5.7-5.8) was run on the 52-week-high signal — the one that *lost* money. Short-term reversal was this project's one positive empirical finding (Section 5.3: "the only signal that actually worked, gross and net of costs," Sharpe 0.22 net on NSE), and it was never re-examined for the exact same uncontrolled-beta artifact that turned out to fully explain the 52-week-high signal's losses. This milestone applies the identical CAPM-style test (*investigations/momentum_reversal_beta.py*) to **both** remaining signals — 12-1 momentum and short-term reversal — long leg, short leg, and combined book, both markets, both frequencies: 24 alpha tests in total.

Short-term reversal's "edge" does not survive. Not one of its 12 alpha tests is significant at conventional levels (all $p > 0.10$, most $p > 0.2$). The positive Sharpe reported all the way back in Section 5.3 was — like the 52-week-high signal's loss — a mix of market-beta exposure and noise, not a demonstrated stock-selection edge. **This project's one previously "positive" empirical result is retracted along with the negative one.**

12-1 momentum tells a genuinely different story. On NSE, momentum shows essentially no significant alpha either (one marginal hit at the 10% level). But on the **US mirror, momentum's long leg and**

combined book show real, statistically robust, largely beta-independent alpha:

	US long leg	US combined book
Daily alpha (annualized)	+8.1%/yr, $p < .0001$	+12.1%/yr, $p = .004$
Monthly alpha (annualized)	+9.2%/yr, $p < .0001$	+15.3%/yr, $p < .0001$
Beta	~+1.0 to +1.1	~-0.05 to -0.24, mostly not significant

The combined long-short book's beta is close to zero and, at daily frequency, not statistically different from zero ($p = 0.38$) — this book is close to genuinely market-neutral **and** has a large, highly significant positive average return. Unlike the marginal, scattered hits dismissed as noise in Section 5.6's multiple-testing discussion, this is a **coherent cluster**: the same signal, the same market, the same direction, significant at the 1% level or better across both frequencies and both the long leg and the combined book — a qualitatively different, much less noise-like pattern than an isolated $p \approx 0.03$ hit.

HOW HARD TO LEAN ON THIS, AS UNDERSTOOD AT THE TIME

This is the strongest, most credible finding in the entire project — but it is one market (the US Kaggle mirror, snapshot ending 2017-11-10), it does not replicate on NSE, and it has not been checked for the specific decay risk this guide's own limitations list has flagged since Part III: momentum was published by Jegadeesh & Titman in 1993, and momentum's premium is well documented in the literature to have weakened somewhat post-publication. Whether this specific alpha holds up in a pre-1994 vs. post-1994 sub-sample split has **not yet been tested** — checked next.

5.10 Milestone 9 — Has US momentum's alpha decayed since publication?

Jegadeesh & Titman published the 12-1 momentum anomaly in the *Journal of Finance* in March 1993. A large literature (notably McLean & Pontiff, "Does Academic Research Destroy Stock Return Predictability?", *Journal of Finance*, 2016) documents that anomaly returns tend to shrink — by roughly 26% after working-paper circulation and ~58% on average after formal publication — once traders can crowd into a known effect. This milestone (*investigations/momentum_publication_decay.py*) splits Section 5.9's long leg and combined-book regressions at **1994-01-01** (~1 year after publication) and re-runs each half separately: 251 pre-1994 rebalances vs. 286 post-1994, out of 537 total.

	Long leg, pre-1994	Long leg, post-1994	Combined, pre-1994	Combined, post-1994
Daily alpha (ann.)	+9.8%/yr, $p = .003$	+6.5%/yr, $p = .032$	+16.1%/yr, $p = .011$	+8.6%/yr, $p = .116$ (n.s.)
Monthly alpha (ann.)	+11.4%/yr, $p = .0004$	+7.1%/yr, $p = .041$	+18.6%/yr, $p = .0008$	+13.1%/yr, $p = .035$

Real, partial decay — exactly the textbook pattern, not full disappearance. Every cut shows the alpha shrinking after 1994: the long leg's daily alpha falls by roughly a third (9.8% to 6.5%/yr) and its monthly alpha by roughly two-fifths (11.4% to 7.1%/yr); the combined book decays by 30-47% depending

on frequency. That magnitude lines up closely with McLean & Pontiff's average post-publication effect across anomalies generally — this isn't an unusually large or suspicious decay, it's the expected one. **The alpha survives in three of four cuts** (long leg, both frequencies; combined book, monthly) but **loses statistical significance in one** (combined book, daily, $p=0.116$). The combined book's beta also drifts from indistinguishable-from-zero pre-1994 (genuinely market-neutral) to weakly negative post-1994 — a secondary sign that the strategy's risk profile itself shifted once the anomaly became public knowledge.

THE HONEST, FINAL-SIZED NUMBER

This project's one durable finding is now stated more precisely: momentum's long leg (and, less robustly, the market-neutral combined book) continued to carry statistically significant, economically smaller alpha through 2017 — not an un-decayed anomaly. That is a real, still-standing finding — three of four regression cuts remain significant in the post-1994 half alone, more than 20 years after publication — but a meaningfully weaker one than Section 5.9's full-sample numbers suggested on their own. **Size any position on the post-decay number, not the full-sample one** — the pre-publication half of the sample describes a market that no longer exists.

UPDATE FROM SECTION 5.11

Section 5.11 re-tested this exact post-1994 number with an actual out-of-sample hedge instead of an in-sample regression, and found it does not survive. Read the box below as the current, superseding answer to "is this decayed number real".

5.11 Milestone 10 — Does the post-1994 alpha survive an actual out-of-sample hedge?

Section 5.10's decay split is a real improvement over a full-sample average, but it has the same limitation Section 5.7's beta regression had before Section 5.8 closed the loop: it fits one beta **in-sample**, using the whole post-1994 sub-sample's own data, then asks whether the average residual differs from zero. A real fund cannot do that — it has to estimate beta from only trailing history and hedge forward. This milestone (*investigations/momentum_hedged_decay_backtest.py*) reuses Section 5.8's rolling, out-of-sample beta hedge (re-estimated every monthly rebalance from only the preceding 252 trading days, applied forward, never looking ahead) on the momentum long leg and combined book, then applies the identical 1994-01-01 split to the resulting **hedged** daily return series.

	Long leg, pre-1994	Long leg, post-1994	Combined, pre-1994	Combined, post-1994
Hedged ann. return	+9.62%/yr	+3.29%/yr	+7.93%/yr	-0.31%/yr
Daily alpha (HAC)	$p=.0010$	$p=.1485$ (n.s.)	$p=.0244$	$p=.5900$ (n.s.)
Monthly alpha (HAC)	$p=.0017$	$p=.1886$ (n.s.)	$p=.0506$ (n.s.)	$p=.5049$ (n.s.)

This is a further, sharper correction, not a confirmation of Section 5.10's "survives in three of four cuts" framing. Once beta is estimated the way a real fund would have to estimate it — from trailing data

only, re-hedged every month, never fit on the same period being tested — the post-1994 alpha is **not statistically distinguishable from zero in either leg**, at either frequency. The combined book's hedged post-1994 average return is outright negative. Pre-1994, the same methodology strongly confirms alpha in both legs ($p \leq 0.025$ in three of four cuts), so the hedge itself isn't simply too noisy to detect a real effect when one is present — it detects it clearly pre-1994 and finds nothing post-1994. The gap between Section 5.10's "three of four cuts survive" and this section's "none survive" is entirely explained by the in-sample vs. out-of-sample distinction: an in-sample regression can fit the specific quirks of the post-1994 data it's being tested against, while a rolling hedge estimated only from prior data cannot.

THE OUT-OF-SAMPLE-CONFIRMED ANSWER, AS FAR AS THIS SECTION TOOK IT

US 12-1 momentum's alpha was real and strong before 1994 and has **not** been demonstrated to survive, in a form an actual fund could have traded, in the more-than-two-decades since. This project's single most credible candidate for a genuine, durable edge does not clear the bar once tested with the same standard of rigor — an actual rolling out-of-sample hedge, not a static or per-era regression coefficient — that Section 5.8 already established as this project's own required standard. This doesn't mean Sections 5.9 and 5.10's findings were wrong for their samples: the full-sample and pre-1994 alpha are both real and robust. It means the finding cannot currently be sized as a forward-looking edge without further work.

UPDATE FROM SECTION 5.12

Section 5.12 investigated this null result in depth, at explicit request, rather than stopping here: is it really just the 2009 momentum crash, or an artifact of the hedge's rolling window? Neither. Read on for the deeper, still-standing answer.

5.12 Milestone 11 — Is the post-1994 null result a crash artifact, a hedge artifact, or genuine decay?

Section 5.11 left two objections explicitly untested: is the post-1994 null result concentrated in a specific regime — most plausibly the well-documented 2009 "momentum crash" (Daniel & Moskowitz, 2016, *Review of Financial Studies*, where past losers momentum strategies were underweighting rebounded violently in the 2008-crisis recovery) — or is it an artifact of the specific 252-trading-day rolling hedge window Sections 5.8 and 5.11 happened to use? This milestone (*investigations/momentum_decay_regime_analysis.py*) runs three checks on the post-1994 hedged return series to answer both.

Check 1 — sub-period breakdown. Splitting post-1994 into five multi-year eras shows a declining trend, not a single bad episode with a recovery: the long leg's hedged excess return runs +6.5%, +10.5%, +3.4%, -3.7%, then **+0.1%** in 2010-2017 — the most recent nine years show essentially zero excess return, well after the 2009 crash was over. The combined book shows the same pattern, ending at **-5.1%** in 2010-2017.

Check 2 — explicit crash-window exclusion (March-August 2009). The crash window itself was severe — the combined book lost 40.2% cumulatively in those 128 trading days alone. Excluding it: the long leg's daily alpha p-value improves from 0.149 to a borderline 0.095 (the crash meaningfully hurt this leg's result), but the combined book's p-value only improves from 0.590 to 0.334, nowhere near significance. The crash is a real contributing factor for the long leg, but does not explain the combined book's null result, and the continued weakness through 2010-2017 shows the effect is not fully explained by a single 2009 episode either way.

Check 3 — hedge rolling-window robustness (126d / 252d / 378d). Re-running the entire hedge with three different lookback windows: pre-1994 alpha is strongly significant at every window tested ($p \leq 0.017$, long leg $p \leq 0.001$) and post-1994 alpha is non-significant at every window tested (p ranges 0.14-0.66 across both legs and all three windows). The null result is not an artifact of the specific 252-day window.

THE DEEPER, STILL-STANDING ANSWER

This deeper investigation does not reverse Section 5.11's finding — if anything it strengthens it. The 2009 momentum crash was real and severe and meaningfully affected the long leg's result, but it is not sufficient on its own to explain either leg's post-1994 null result, and the hedge's design choice is not driving it either. **The era-by-era trend (positive and shrinking through the 2000s, negative through the crisis, and still flat or negative for the nine years since) is the signature of genuine, ongoing decay rather than one bad shock the strategy simply hasn't yet recovered from.** A finding that survives a genuine attempt to explain it away deserves more confidence, not less.

UPDATE FROM SECTION 5.13

Section 5.13 asks whether this whole pattern is unique to momentum, by applying the same toolkit to the other two US signals. The answer splits: one signal never had genuine alpha at all; the other shows the identical decay signature as momentum.

5.13 Milestone 12 — Is "real pre-1994, decayed since" specific to momentum, or market-wide?

Sections 5.10-5.12 built a specific toolkit — a rolling, out-of-sample beta hedge split at 1994-01-01 — and applied it only to momentum, because momentum was the one signal with significant full-sample alpha worth investigating. But the 52-week-high and short-term reversal signals were both declared dead using a *full-sample* beta-adjusted regression (Sections 5.7-5.9), which would hide the exact same pattern found for momentum: real, significant alpha in an early era, averaged down to statistical noise by a decayed later era. This milestone (*investigations/all_signals_decay_analysis.py*) applies the identical pre/post-1994 hedge to **all three** US signals — not to re-answer an already-settled question (NSE momentum's full-sample null was already established in Section 5.9, so re-running the hedge there would add nothing), but to ask a genuinely new one: is the decay pattern unique to momentum, or a broader feature of the US market?

Signal / leg	Pre-1994 ann. ret / p	Post-1994 ann. ret / p
12-1 momentum, long leg	+9.62%/yr, p=.001	+3.29%/yr, p=.149
12-1 momentum, combined	+7.93%/yr, p=.024	-0.31%/yr, p=.590
52-week-high, long leg	-0.21%/yr, p=.704	-2.06%/yr, p=.468
52-week-high, combined	-8.46%/yr, p=.776	-6.68%/yr, p=.401
Reversal, long leg	+5.31%/yr, p=.046	+0.25%/yr, p=.629
Reversal, combined	-0.41%/yr, p=.293	-0.57%/yr, p=.576

Two genuinely different stories, not one. The 52-week-high signal shows **no significant alpha in either era, in any leg** — confirming Sections 5.7-5.8's conclusion that this signal never had genuine stock-selection skill in either direction; its full-sample "loss" was uncontrolled beta from the start, not a decayed edge. **Short-term reversal's long leg tells a different story: it shows the identical decay pattern as momentum** — a real, statistically significant pre-1994 alpha (+5.31%/yr, p=0.046) that decays completely to noise post-1994 (+0.25%/yr, p=0.629). This was invisible in Section 5.9's full-sample regression, which averaged the genuine early effect with the decayed later one and correctly found no full-sample significance — but "no full-sample significance" is not the same claim as "never had a genuine edge," and this milestone shows reversal's long leg did.

TWO SIGNALS, ONE SHARED STORY

Section 5.9's headline claim — that reversal's "edge" did not survive the same scrutiny applied to the negative one — is accurate for the full-sample regression it ran, but incomplete: reversal's long leg was never pure noise, it was a real, decayed effect exactly analogous to momentum's, just never tested with the era-split methodology that only existed starting at Section 5.10. **The "real pre-1994, decayed since" pattern is not a momentum-specific quirk — it appears in two of this project's three signals' long legs (momentum and reversal) and is absent from the third (52-week-high, which never had genuine alpha at all)**, consistent with a market-wide explanation — the same 1990s-2000s scaling-up of quantitative, cross-sectional equity strategies this project's own case studies (LTCM, the 2007 Quant Quake) already document as having transformed US equity markets over exactly this period — rather than an idiosyncratic property of momentum alone.

UPDATE FROM SECTION 5.14

Section 5.14 asks a sharper question: not just whether there was a decay, but exactly how fast, and when it actually crossed zero — quantified directly, instead of assumed from a single arbitrary cutoff date. The answer reshapes how momentum's story should be told.

5.14 Milestone 13 — Quantifying the decay: a continuous rate, not a binary 1994 cut

Every decay test so far (Sections 5.10-5.13) used a single, somewhat arbitrary binary split — 1994-01-01, chosen as “~1 year after Jegadeesh & Titman's 1993 publication.” That answers “was there a difference before vs. after this one date,” not “how fast did the edge erode, or when did it actually run out.” This milestone (*investigations/decay_rate_estimation.py*) quantifies the decay directly: for each signal's out-of-sample-hedged long leg, over the *full* sample (no pre/post split), it regresses $\text{hedged_return} = \alpha + \text{slope} \times (\text{years since hedge coverage began})$, with HAC standard errors. *Slope* is a direct, continuously-estimated annual decay rate with its own p-value, and (α , slope) together imply a zero-crossing date.

Signal	alpha (start, ann.)	slope (change/yr)	Zero-crossing
12-1 momentum, long leg	+12.65%/yr, p=.008	-0.23%/yr, p=.152 (n.s.)	ill-conditioned
Reversal, long leg	+13.46%/yr, p=.014	-0.41%/yr, p=.026	2004-11-22

Momentum's decay is not a smooth line — it does not even pass as one. The full-sample linear-trend slope for momentum's long leg is *not* statistically significant ($p=0.15$ daily, $p=0.25$ monthly): a single straight line across the whole 1972-2017 hedged return series is a poor fit, not because the effect didn't decay, but because it didn't decay *smoothly*. A 5-year rolling-window trajectory shows why: annualized hedged returns stay consistently strong (roughly +4% to +22%/yr, noisy but never close to zero) all the way from the 1970s through the window ending January 2008 — including every year of the supposed “post-1994 decay” period — and only then drop sharply, turning negative for every rolling window from 2009 onward. **Splitting explicitly at September 2008 instead of January 1994 gives a far cleaner separation:** pre-Sept-2008 long-leg alpha is +8.02%/yr, $p=0.0005$ ($n=8,983$ days) vs. post-Sept-2008 alpha of -0.56%/yr, $p=0.998$ ($n=2,309$ days, utterly indistinguishable from zero) — a starker divide than the 1994 split produced ($p=0.001$ pre-94 vs. $p=0.149$ post-94). This refines, rather than contradicts, Sections 5.10-5.12's momentum finding: the direction was right, but describing it as gradual, publication-driven decay since 1994 is not well supported by the data's actual shape. A sudden regime shift around the 2008-09 financial crisis — plausibly the same 2009 momentum crash examined directly in Section 5.12, or the post-crisis scaling-up of quantitative strategies — is a better-supported story for *when and how* momentum's edge disappeared than slow 1990s crowding.

Reversal's decay is closer to the smooth story the publication-decay literature predicts. Its full-sample linear trend *is* statistically significant (slope -0.41%/yr daily, $p=0.026$; -0.39%/yr monthly, $p=0.027$), with an implied zero-crossing around November 2004. The rolling trajectory confirms a genuine decline, though a real one, not a perfectly straight line: annualized returns fall from a very strong +27%/yr in the late 1970s to negative territory by the mid-1980s, stay negative through the 1990s, then show an unexplained recovery bump (+6% to +14%/yr) from 2000-2004 before declining again through the 2010s. The overall downward trend is real and significant, but “smooth, monotonic decay” oversimplifies a pattern that includes a multi-year partial recovery in the middle.

RIGHT DIRECTION, WRONG SHAPE

Quantifying the decay rate, rather than assuming a fixed cutoff, shows the two signals' declines have genuinely different shapes. Reversal's long leg decays roughly the way the publication-decay literature describes — a real, continuous, statistically significant downward trend, crossing zero around 2004. Momentum's long leg does not: it held essentially flat and strong for 35+ years and then broke sharply around the 2008-09 financial crisis, a pattern a continuous linear-decay model fits poorly and a structural-break framing fits well. **Treat momentum's weakness as "broke around 2008-09," not "has been gradually decaying since 1994" — the earlier milestones' binary split happened to land on the right side of the qualitative story without correctly identifying its shape or timing.**

UPDATE FROM SECTION 5.15

Section 5.15 formalizes the "broke around 2008-09" claim with an actual structural-break test instead of a descriptive comparison — and the formal result is more conservative than this section's own framing suggested.

5.15 Milestone 14 — A formal structural-break test, not a descriptive comparison

Section 5.14's "split at September 2008 instead of 1994" comparison was descriptive: the break date was chosen *after* looking at momentum's rolling-window trajectory, and testing "the best of many candidate split dates" is expected to look impressive even under a null of no true break — a comparison, not a test. This milestone (*investigations/structural_break_test.py*) runs two properly specified tests instead.

Method A — a literature-motivated Chow-style test at a single, pre-specified date (2008-09-01, motivated externally by Daniel & Moskowitz's (2016) documented 2009 momentum-crash episode, not by inspecting this project's own plot). A single pre-specified date needs no multiple-testing correction.

Method B — a Quandt-Andrews-style sup-Wald test that does *not* assume a break date. It searches every candidate date in the central 70% of the sample (15% trimmed from each end, standard Andrews (1993) practice), computes the HAC-robust level-shift statistic at each one, and takes the maximum — both its value and which date it occurs at. Because searching many candidates inflates the false-positive rate of a naive comparison, this project's sandboxed environment (which cannot fetch Andrews' published critical-value tables) instead builds its own empirical null: fit a no-break model to the real data, block-bootstrap its residuals (400 draws, 12-month blocks, preserving autocorrelation), rerun the full candidate search on each synthetic no-break series, and compare the real maximum statistic to that simulated null distribution.

	Momentum	Reversal
Method A: shift at 2008-09, daily	-9.31%/yr, p=.026	-4.02%/yr, p=.337
Method A: shift at 2008-09, monthly	-9.10%/yr, p=.052	-3.90%/yr, p=.316

	Momentum	Reversal
Method B: data-driven break date	2010-12-31	1980-08-29
Method B: sup t (bootstrap p)	2.771 (p=.138, n.s.)	3.369 (p=.048)

Momentum: the specific 2008 hypothesis holds; an unconstrained search does not decisively confirm it as the dominant break. Method A confirms a real, statistically significant level shift specifically at the literature-motivated date ($p=0.026$ daily). But Method B's unconstrained search finds its single best-fitting break at **December 2010, not September 2008** — 27 months later — and that best-fit statistic is *not* significant once corrected for having searched ~375 candidate dates (bootstrap $p=0.138$). This nuances Section 5.14's framing: there is good evidence of weakening tied specifically to the 2008-09 crisis window, but the data does not decisively pin down one single, dominant structural break — a specific, externally-motivated hypothesis survives; an unconstrained "find the best break anywhere" search does not clearly beat noise.

Reversal: no evidence for a 2008-tied break — but a genuine, significant break turns up much earlier, around 1980. Method A finds nothing at 2008 (expected: Section 5.14 already showed reversal's decline predates the financial crisis by decades). Method B's unconstrained search finds a significant break (bootstrap $p=0.048$) at **August 1980** — the early, sharp decline from the extraordinarily high +27%/yr level of the late 1970s that Section 5.14's rolling trajectory showed, not a smoothly accumulating multi-decade slope. This further refines Section 5.14's "smooth, statistically significant linear decay" framing for reversal: part of that significant linear trend is better described as one sharp early-1980s adjustment than continuous decay across the whole sample.

THE FORMAL, MORE CONSERVATIVE ANSWER

Neither signal's decline is best described as either a perfectly smooth trend or a single obvious break once tested formally and correctly for multiple-testing bias. Momentum shows a real, hypothesis-confirmed weakening at the 2008-09 crisis, but not an unambiguous single structural break when searched for without that prior. Reversal shows a genuine, statistically significant break, but decades earlier than any milestone had proposed, complicating rather than confirming the smooth-decay story. **Both results are more conservative and more precisely qualified than any earlier milestone's framing** — consistent with this project's repeated experience that the more rigorous test usually narrows, rather than confirms, the simpler story that preceded it.

UPDATE FROM SECTION 5.16

Section 5.16 asks WHY reversal broke around August 1980, at explicit request, rather than reporting the date and moving on. The answer is a major correction, not a refinement: reversal's entire positive finding, carried since Section 5.10, turns out to be an artifact of unreliable early data.

5.16 Milestone 15 — What actually happened in 1980? A major correction

Section 5.15 found a genuine, statistically significant structural break for reversal around August 1980, but offered no explanation for the date. At explicit request, this milestone (*investigations/reversal_1980_break_diagnostics.py*) investigates the mechanism directly — and the result substantially revises, rather than merely refines, this project's understanding of reversal's status.

The universe is severely thin in the 1970s-early 1980s — 2-4 stocks, not a portfolio. From 1972 through 1983, reversal's decile "long leg" contains only **2-4 stocks**, drawn from a total universe of just 9-14 names. The underlying universe itself is a small, hand-picked subset of *today's* largest surviving companies (AAPL, JPM, JNJ, PG, XOM, KO, PEP, WMT, HD, DIS, CVX, PFE, INTC, VZ, T, MRK, ABT, MCD) backfilled to their earliest available data — a textbook survivorship-biased sample: every name in this era's universe is, by construction, a company that *did* go on to become a mega-cap winner by 2017. Any stock that dipped temporarily in the 1970s-80s was, in hindsight, virtually guaranteed to "bounce back," mechanically inflating an apparent reversal effect that has nothing to do with genuine investor overreaction.

A genuine data-quality defect, not previously flagged, compounds the problem. Scanning this era for single-day moves exceeding 50% (the usual fingerprint of an unadjusted stock split) turns up two real anomalies: **WMT** drops 52% on 1974-12-06 and then jumps back +109% twelve days later on 1974-12-18 (a round-trip, consistent with a split-adjustment error that later self-corrects in the raw feed), and **INTC** jumps +101% on 1972-01-27 and a further +51% on 1972-06-22. These are new, previously undocumented data-quality issues in the US Kaggle mirror.

Start date	Reversal ann. ret / p	Momentum (control) ann. ret / p
1972-01-01 (full sample)	+2.64%/yr, p=.068	+6.21%/yr, p=.0008
1978-01-01	-0.59%/yr, p=.833	+6.38%/yr, p=.0014
1980-08-29 (break date itself)	-1.24%/yr, p=.922	+6.20%/yr, p=.0027
1985-01-01	-1.20%/yr, p=.935	+5.90%/yr, p=.0056
1990-01-01	-0.73%/yr, p=.896	+5.47%/yr, p=.0192
1995-01-01	+0.48%/yr, p=.581	+3.75%/yr, p=.121

Reversal's entire positive-alpha claim depends on the unreliable 1972-1977 window and disappears completely once it is excluded — from any start date at or after 1978, including the Method B break date itself, p ranges 0.54-0.96 and the point estimate is usually slightly negative. **Momentum, run as a control through the exact same thin, survivorship-biased, partly data-glitched early universe, is unaffected:** it remains highly significant ($p \leq 0.02$) at every start date through 1990, fading only toward the already-established, externally-motivated 2008-09 territory from 1995 onward. The same data limitation affects both signals identically; only reversal's finding depended on it.

RETRACTED, NOT REFINED

The August 1980 "break" is not evidence of a real 1980 market event: it is the Quandt-Andrews search correctly detecting that reversal's entire apparent edge lives inside an unreliable, thin, survivorship-biased, and partly data-glitched early sample window, with nothing genuine on either side of any split point once that window is excluded. **The "reversal has genuine pre-2005 alpha" claim carried since Section 5.10 through Section 5.15 should be treated as retracted, not merely decayed or narrowed: reversal shows no reliably demonstrated edge anywhere in this dataset once the unreliable years are excluded.** Momentum is unaffected by this correction and, if anything, comes out more strongly validated: the same diagnostic that broke reversal's finding left momentum's intact.

UPDATE FROM SECTION 5.17

Section 5.17 asks whether momentum's confirmed 2008-09 break has a real, named mechanism behind it, not just a date — and reuses this project's own crash-risk toolkit, rejected for a different signal, to test it.

5.17 Milestone 16 — Does the classic momentum-crash mechanism explain the 2008-09 break?

Sections 5.13-5.14 confirmed momentum's long leg broke sharply around September 2008 but left the *mechanism* unexplained — Section 5.12 checked a fixed calendar window (March-August 2009) and found it only partially mattered. This milestone (*investigations/momentum_crash_mechanism_2008.py*) tests the actual, named mechanism instead of a calendar window: the classic Daniel & Moskowitz (2016) momentum-crash trigger (past losers snapping back hard in a high-volatility market rebound), using the same look-ahead-free Bear × High-Volatility regime-interaction regression this project built and validated in Sections 5.7-5.8 — where it was tested on the 52-week-high signal, full-sample, and found *not* significant. Applying the identical regression to momentum's out-of-sample-hedged long leg, split at the same literature-motivated 2008-09-01 date used since Section 5.14:

	Pre-2008-09	Post-2008-09
Bear x High-Vol regime frequency	6.1% of days	9.1% of days
Bear x High-Vol interaction coef.	-0.00062/day, p=.424 (n.s.)	-0.00218/day, p=.0081
Baseline (non-crash-regime) alpha	+0.00038/day, p=.003 (~+9.6%/yr)	+0.00017/day, p=.203 (n.s.)
Implied return, Bear+High-Vol day	~-11%/yr (n.s.)	~-37%/yr

The crash mechanism was dormant before 2008 and activated after. Pre-2008-09, the Bear × High-Vol interaction is small and not remotely significant (p=0.42) — consistent with Sections 5.7-5.8's original finding that momentum-crash risk wasn't a real driver of this project's data. Post-2008-09, the identical interaction term becomes large, negative, and statistically significant (p=0.008): on the roughly

9% of post-2008 trading days that are both high-volatility and in a trailing bear-market state, momentum's long leg loses at an annualized rate around 37%, while its baseline (non-crash) daily alpha has fallen to statistically indistinguishable from zero. The combined long-short book shows the same sign and similar magnitude but weaker significance ($p=0.21$) — consistent with the extra noise from a still-volatile short leg diluting the signal.

A MECHANISM, NOT JUST A DATE

This gives momentum's 2008-09 structural break a genuine causal mechanism, not just a confirmed date. The classic momentum-crash dynamic this project rejected in Sections 5.7-5.8 (for the 52-week-high signal, full-sample) and that Section 5.12 found only partially relevant to a fixed 2009 calendar window, turns out to be real for momentum specifically — but **conditional on era: dormant through the pre-crisis decades, then active since**. This is consistent with, and adds mechanistic detail to, the broader post-2008 quant-crowding story already touched on in this project's own Quant Quake case study — a market where momentum-following capital had scaled up enough by 2008 for the classic crash dynamic to actually bite.

UPDATE FROM SECTION 5.18

Section 5.18 asks whether the same crash mechanism, and a more rigorous re-test of NSE momentum itself, hold up on the market this project has tested the most and trusted the least.

5.18 Milestone 17 — Does the momentum-crash mechanism replicate on NSE?

Section 5.17's finding was established on one market. This milestone (*investigations/momentum_crash_mechanism_nse.py*) tests it on NSE (India) — a market that lived through the same 2008 global crisis. Before trusting anything here, the Section 5.16 lesson applies: NSE momentum's decile long leg holds 6-10 names throughout 2000-2021 (vs. the US mirror's notorious 2-4 names in the 1970s-80s), a reasonable portfolio size, not the thin-universe artifact that sank reversal.

A genuine methodological gap, closed first. Section 5.9 tested NSE momentum's alpha with a single *static* full-sample beta regression and found no significant result. That is a materially cruder test than the rolling, out-of-sample hedge this project built in Section 5.8 and has used for every US momentum test since — but it had never been applied to NSE momentum before. Doing so now:

	Long leg, daily	Long leg, monthly
Full sample (2000-2021)	+3.56%/yr, $p=.170$ (n.s.)	+4.55%/yr, $p=.168$ (n.s.)
Pre-2008-09	-2.82%/yr, $p=.744$ (n.s.)	-1.70%/yr, $p=.754$ (n.s.)
Post-2008-09	+7.66%/yr, $p=.044$	+7.94%/yr, $p=.073$ (marginal)

Full-sample, the more rigorous hedge reaffirms Section 5.9's conclusion — no significant NSE momentum alpha, daily or monthly, now on firmer methodological footing. But a **post-2008-specific signal emerges** that the cruder full-sample test could not have seen: daily-frequency alpha is significant ($p=0.044$), monthly is marginal ($p=0.073$). This is genuinely new, but should be read as "promising, not confirmed" — this project's own standard label for exactly this strength of evidence (Section 5.10) — and comes with a real caveat: NSE's universe itself grew from ~30 to 48 names over this window, so part of the post-2008 improvement in signal strength may reflect a less thin, better-populated cross-section rather than a genuine change in the underlying economics.

The crash-specific mechanism itself does not replicate. Applying Section 5.17's exact Bear $\times$ High-Vol regression to NSE: the interaction term is never significant ($p=0.96$ full-sample, $p=0.34$ pre-2008, $p=0.66$ post-2008) — unlike the US, where it activated sharply post-2008. What NSE *does* show is a plain, unconditional **Bear** effect: the long leg loses significantly in any trailing bear market (coef=-0.00095/day, $p=0.002$ full-sample; $p=0.001$ post-2008), regardless of whether volatility is simultaneously high. That is a related but mechanistically different pattern from the US's specific *crash-rebound* dynamic — NSE momentum looks bear-market-sensitive in general, not crash-rebound-sensitive in particular.

A NEW SIGNAL, AND A MECHANISM THAT DOESN'T TRAVEL

This is not a repeat of the already-settled "does NSE momentum work" question — it is two new, honestly-qualified findings. The specific momentum-crash mechanism confirmed for the US in Section 5.17 is not universal; it does not show up in NSE the same way, even though NSE lived through the same 2008 crisis. Separately, a more rigorous re-test surfaced a genuinely new, if still tentative, post-2008 NSE momentum signal that a cruder test had missed — evidence that this project's own methodology upgrades are still capable of finding things earlier, less careful passes did not, even on a market already checked multiple times.

UPDATE FROM SECTION 5.19

Section 5.17's "post-2008" block pools the 2008-09 crisis with eight subsequent years. Section 5.19 splits it and asks whether the crash-regime effect actually persisted, or was the crisis alone.

5.19 Milestone 18 — Does the 2008-09 crash mechanism persist after the crisis?

Section 5.17 tested the Bear $\times$ High-Volatility interaction on the *whole* post-2008-09 block (Sept 2008 through the end of the US mirror's coverage in Nov 2017, $n=2,309$ days) and found it significant ($p=0.008$), reading that as the crash mechanism having "activated" for good after 2008. But that block pools two very different periods: the acute 2008-09 crisis itself, and eight subsequent years. This milestone (*investigations/momentum_crash_mechanism_persistence.py*) splits the post-2008-09 block at 2009-12-31 and re-runs the identical regression on each half separately.

The split immediately surfaces a structural fact this project had not checked before: **the US mirror's trailing-12-month market return never went negative again after September 2009**, all the way through the end of its coverage in November 2017 — the 2011 European-debt-crisis selloff and the Aug 2015-Feb 2016 drawdown were sharp but both recovered within the 252-day lookback before ever registering as a sustained bear state under this project's own (Section 5.5) definition.

	Pre-2008-09	Crisis (2008-09 to 2009-12)	Post-crisis (2010-01+)
Trading days	8,983	337	1,972
Bear x High-Vol frequency	6.1%	62.3%	0.0%
Interaction coef (long leg)	-0.00062/day, $p=.42$ (n.s.)	+0.00115/day, $p=.26$ (n.s.)	cannot be estimated
High-Vol alone (crisis)	-	-0.00361/day, $p<.0001$	-
Bear alone (crisis)	-	-0.00188/day, $p=.036$	-

Two refinements to Section 5.17's framing, not a retraction of its core fact. Momentum's long leg still lost heavily during 2008-09 — that empirical fact is unchanged and confirmed again here. But two things Section 5.17's pooled test could not show on its own:

- **There is no evidence of a persisting post-2008 bear-market regime to test.** Because the trailing-return bear flag never fired again after September 2009, the "post-2008-09" window Section 5.17 tested is, for the interaction term's purposes, entirely carried by the 16-month crisis sub-window — the other ~2,000 days post-crisis contribute zero Bear+High-Vol observations. Framing this as a standing "post-2008 regime change" was an overstatement; the honest description is "active during the 2008-09 crisis, untested since, because no comparable bear market has recurred in this sample."
- **Isolated to the crisis window alone, the specific multiplicative interaction is not what's doing the work.** Run on the crisis's 337 days by themselves, the interaction term is *not* significant ($p=0.26$ long leg, $p=0.31$ combined) — instead, high volatility alone ($p<0.0001$) and the bear-market flag alone ($p=0.036$ long leg, $p=0.005$ combined) each independently explain the losses. Section 5.17's pooled significance for the interaction specifically came from contrasting the crisis window against ~2,000 calm, non-bear post-crisis days as an implicit control group — a valid test of "did something change after 2008," but weaker evidence for "the interaction, specifically, is the mechanism" than the $p=0.008$ headline number suggested in isolation.

Neither sub-window shows any hedged-alpha significance in the plain daily/monthly check either (crisis: $p=0.84$ daily, $n=15$ months insufficient for HAC; post-crisis: $p=0.86$ daily, $p=0.57$ monthly) — consistent with Section 5.17's original post-2008 baseline-alpha result ($p=0.20$, n.s.).

ONE CRISIS, NOT YET A REGIME

The 2008-09 break is real and tied to volatility and bear-market state, but this project has no evidence it is a *standing* feature of the post-2008 world, because the specific bear-market condition this project uses to define "crash regime" has not recurred since 2009 in this sample. A risk model built on Section 5.17's finding alone would be over-claiming "momentum has been crash-exposed since 2008"; the accurate claim is narrower: "momentum was crash-exposed during the one bear market this sample's post-2008 window actually contains." Whether the mechanism would reactivate in a future bear market is a hypothesis this data cannot confirm or rule out — there simply hasn't been one to test against since.

UPDATE FROM SECTION 5.20

Section 5.19's central claim rests on a 252-day bear-market lookback. Section 5.20 asks whether that claim, and Section 5.17's pattern, survive other reasonable choices of that window.

5.20 Milestone 19 — Does the crash-mechanism finding survive different regime-construction choices?

Every crash-regime result since Section 5.6 rests on two parameter choices baked into *build_regime_dummies()*: a 21-trading-day realized-volatility window and a 252-trading-day (~1-year) trailing-return lookback for the "Bear" flag. Neither had been stress-tested. This matters more than usual for Section 5.19's finding specifically, because "no bear market recurred after 2009" is a claim that can only be as robust as the lookback window that defines "bear market" — a shorter window would register the sharp 2011 and 2015-16 drawdowns that a 1-year lookback smooths away. This milestone (*investigations/momentum_crash_regime_robustness.py*) reimplements the regime construction with configurable windows (the original function used by every prior milestone is untouched) and sweeps a 4×4 grid: volatility windows of 10, 21, 42, and 63 trading days, crossed with bear-market lookbacks of 126, 189, 252, and 378 trading days — 16 combinations, each checked two ways: does a Bear regime ever fire after 2010-01-01, and does Section 5.17's "interaction dormant pre-2008-09, significant post-2008-09" pattern still hold.

	Bear lookback 126d	189d	252d (default)	378d
Bear regime fires post-2010?	Yes (200 days)	Yes (47 days)	No	No
Sec. 5.17 pattern replicates	1 of 4 vol windows	2 of 4	2 of 4 (incl. default)	3 of 4

Section 5.19's "no bear regime since 2009" is not robust to the lookback window — this is a real qualification, not just a robustness footnote. At the two shorter, equally standard lookbacks (126 trading days ≈ 6 months, 189 ≈ 9 months), the trailing-return bear flag *does* fire after 2010 — 200 days clustered in 2010-11 and 2015-16, and 47 days in the same years respectively — exactly the 2011 European-debt-crisis selloff and the Aug 2015-Feb 2016 drawdown that Section 5.19 already named as sharp-but-short episodes the 252-day window happened to smooth away. Only at lookbacks of 252 days or longer does the "no recurrence" claim hold. Section 5.19's finding should be read as conditional on this

project's own 1-year convention, not as a lookback-independent fact about the market.

Section 5.17's interaction pattern itself replicates in 8 of 16 combinations — real, but concentrated in a specific part of the parameter space. It never holds at the shortest volatility window (10 days, 0 of 4 bear-lookbacks) — at that window the pre-2008 era itself becomes significant in one case, breaking the "dormant before" half of the claim. It rarely holds at the shortest bear-lookback (126 days, 1 of 4 vol-windows). But it holds reliably — 3 of 4 bear-lookbacks — at this project's own default 21-day volatility window and the adjacent 42-day window, and the default parameter combination that Sections 5.17-5.19 actually used sits squarely inside that reliable region, not at an edge case cherry-picked to produce significance.

ROBUST PATTERN, FRAGILE ABSENCE-CLAIM

Two different answers for two different claims sharing the same regime-construction code. Section 5.17's qualitative story — the crash mechanism was quiet before 2008-09 and active during/after it — is reasonably robust to nearby parameter choices, failing only at unusually short windows this project never actually used. Section 5.19's stronger, more specific claim — that *no* bear market recurred anywhere in the post-2009 sample — is fragile: it depends on choosing a bear-lookback of a year or longer, and a 6- or 9-month lookback, an equally defensible convention, tells a different story. The honest combined statement: momentum's crash-regime exposure is confirmed for the 2008-09 crisis under every parameter choice tested, and whether it also showed up in 2011 or 2015-16 is a genuinely open question this project has not yet run — the persistence test only ran under the window where no second bear regime exists to test it against.

UPDATE FROM SECTION 5.21

Section 5.20 closes that open question directly: re-running the persistence test under the lookbacks where a real post-2010 bear regime exists.

5.21 Milestone 20 — Did the crash mechanism reactivate in 2011 or 2015-16?

Section 5.20 left one question explicitly open: at bear-market lookbacks of 126 and 189 trading days (~6 and ~9 months), a bear regime *does* recur post-2010 — clustered in 2010-11 (the European debt-crisis selloff) and 2015-16 (the Aug 2015-Feb 2016 drawdown) — giving 148 and 31 Bear+High-Vol days respectively to actually test the crash mechanism's persistence against, something Section 5.19's default 252-day lookback could not do at all. This milestone (*investigations/momentum_crash_mechanism_recurrence.py*) closes that thread: re-running the Bear × High-Vol interaction regression on the 2010-onward window, under both alternate lookbacks, with everything else unchanged.

	Bear lookback 126d	Bear lookback 189d
Bear+High-Vol days, 2010 onward	148 (7.5% of era)	31 (1.6% of era)

	Bear lookback 126d	Bear lookback 189d
Interaction coefficient	-0.00172/day, p=.203 (n.s.)	-0.00143/day, p=.443 (n.s.)
Bear-alone coefficient	+0.00126/day, p=.226 (n.s.)	+0.00212/day, p=.023

The mechanism did not reactivate, under either alternate lookback — and where a coefficient was significant, it pointed the wrong way for crash risk. At neither window is the Bear × High-Vol interaction anywhere near significant post-2010 ($p=0.20$, $p=0.44$), despite now having real bear-regime days to test it against. At the 189-day lookback the bear-market main effect *is* significant ($p=0.023$) — but positive, not negative: momentum's long leg did somewhat *better*, not worse, during 2010-onward trailing-bear periods, the opposite sign from both the crisis-era coefficient and what the crash-risk mechanism predicts. This is not the pattern of a dormant mechanism waking back up; it is closer to no pattern at all.

TESTED AND NOT FOUND, NOT JUST UNTESTED

This strengthens, rather than merely leaves open, Section 5.19's original framing. The crash mechanism is not just *untested* outside the 2008-09 crisis under this project's default parameters — now that two alternate, equally standard parameter choices supply real bear-regime days post-2010, the mechanism is *tested and not found* there. The honest, now-complete statement across Sections 5.17, 5.19, 5.20, and 5.21: the Bear × High-Vol interaction explains momentum's 2008-09 losses specifically, has not reappeared in either of the two next bear-adjacent episodes this sample contains under any lookback tested, and a standing "momentum is crash-exposed" risk rule would have been wrong to apply in both 2011 and 2015-16. Whether it would reactivate in a genuinely severe future crisis, as opposed to the milder episodes of 2011 and 2015-16, remains open — this sample simply has not contained one since 2009.

UPDATE FROM SECTION 5.22

The crash-mechanism chain is now closed. Section 5.22 turns the same pooled-window lesson back on this project's own other open finding: the tentative NSE post-2008 momentum signal.

5.22 Milestone 21 — Does the tentative NSE signal survive its own named caveat?

Section 5.18's tentative post-2008 NSE momentum signal (long leg, daily $p=0.044$, monthly $p=0.073$) shipped with an explicit, unresolved caveat: NSE's universe grew from ~30 names in 2000 to a full 48 by late 2010, so part of the apparent post-2008 improvement could reflect a less thin, better-populated cross-section rather than a genuine change in the underlying economics. That caveat was named but never tested — until now. Checking the NSE mirror's daily coverage directly finds the universe was still expanding through 2009 (45 of 48 names on average) but has been **perfectly fixed at exactly 48 names, every single day, from 2010-11-04 through the end of the sample (2021-04-30)** — 10.5 of the ~12.7 years in Section 5.18's "post-2008" window. That split gives a clean test: does the signal survive when restricted to only the years where a growth confound is definitionally impossible, because the universe never changed size at all?

	Post-2008 full	Growing (2008-09 to 2010-11)	Stable (2010-11+, fixed 48)
Days	3,114	535	2,579
Daily ann. return / p	+7.66%/yr, $p=.044$	+23.72%/yr, $p=.112$ (n.s.)	+4.60%/yr, $p=.180$ (n.s.)
Monthly ann. return / p	+7.94%/yr, $p=.073$	+21.55%/yr, $p=.132$ (n.s.)	+4.96%/yr, $p=.250$ (n.s.)

Not the confound named, but a different and arguably more serious problem for the same finding.

The growth-confound story predicted the *growing* years would show inflated, thin-universe significance that the *stable* years would lack — but the opposite pattern shows up in the point estimates: the growing sub-period's annualized return is actually *larger* (+23.72%/yr vs. +4.60%/yr), not smaller, so this is not evidence the universe growth specifically inflated the signal. What it does show is something this project has now learned to recognize from Sections 5.19-5.21: **the full-window significance is a pooling artifact**. Neither natural sub-period — not the growing years, not the fully stable, fixed-48-name years — reaches significance on its own, at either frequency. The signal exists only when the two are pooled together.

A NAMED CAVEAT, FINALLY TESTED

Section 5.18's own "promising, not confirmed" label undersold how fragile this finding is. It is not just unconfirmed; it does not survive being split by the one structural feature (universe stability) this project already knew was a live concern, in either direction the split could have gone. Applying the same lesson Sections 5.19-5.21 learned about the crash mechanism — a pooled window's significance can come from combining two periods rather than a persisting effect in either — to this project's own tentative finding rather than only to someone else's: the NSE post-2008 momentum signal should be read as *not currently demonstrated*, not merely as an open, promising thread.

UPDATE FROM SECTION 5.23

Both of this project's remaining open threads are now closed. Section 5.23 opens a new one instead: a genuinely independent third market.

5.23 Milestone 22 — A third, independent market: does momentum replicate on ASX?

NSE and the US Kaggle mirror are this project's only two markets so far, and both trace back to community re-exports of specific existing datasets (NSE: one uploader's CSV; US: a well-known Kaggle "Huge Stock Market Dataset" mirror). Neither is an independent check on whether momentum is a market-wide phenomenon or an artifact of how those two particular datasets happen to be built. This milestone looks for a genuine third source.

Finding one was harder than expected — worth documenting as part of the result, not just a footnote. An extensive search for a European per-company OHLCV mirror (the first choice) found none reachable from this sandbox: *stooq.com*, *huggingface.co*, *github.com*'s own HTML pages, and *api.github.com*'s general repo-browsing API are all blocked by the network proxy here — only *raw.githubusercontent.com* is allowlisted, and only for exact, known file paths. Several candidate European-stock repositories turned out to be pipeline code that fetches from Yahoo Finance or Kaggle at run time (also blocked), not committed price data. The source that finally worked is *grantcarthew/data-asx-historical-share-tables* — a GitHub mirror of the Australian Securities Exchange's own daily S&P/ASX300 constituent report emails, 2009-10-20 to 2015-12-31. Not European, but a genuinely independent developed market: different exchange, different uploader, official ASX report emails rather than a Kaggle re-export, from both NSE and the US mirror.

Applying this project's current best methodology directly, rather than its history. Rather than re-running the original crude full-sample regression Sections 5.7-5.9 started with, this test goes straight to the out-of-sample rolling hedge and HAC significance check this project has used for every market since Section 5.11/5.18:

	Long leg	Combined long-short
Raw ann. return (unhedged)	+8.29%/yr	+27.15%/yr
Hedged daily ann. return / p	+11.14%/yr, p=.0085	+35.45%/yr, p=.0006
Hedged monthly ann. return / p	+9.97%/yr, p=.0141	+30.97%/yr, p=.0001
Mean hedge beta	+0.83	-0.42

Momentum replicates cleanly on ASX — the strongest, most unambiguous result of the three markets tested. Both legs are significant at both frequencies, on a proper out-of-sample hedge from day one (this market never went through the "static full-sample regression first, hedge added later" history NSE and the US mirror did). The combined book's large magnitude has a plausible, checked explanation rather than being a red flag on its own: the short leg's mean beta (-1.26) is far more negative than the long leg's (+0.83) is positive, consistent with Australia's well-known 2011-2015 mining and resources downturn — momentum's short leg would have been loaded with high-beta miners that kept underperforming through exactly that window, and the rolling beta estimates themselves are reasonably stable (std 0.39, no extreme outliers) rather than a symptom of an unstable hedge.

CLEANEST RESULT, SHORTEST HISTORY

A genuine limitation, stated plainly: this is the shortest history of the three markets. Six years (1,501 trading days) is enough for a full-sample significance test but not for the era-splitting, structural-break, or persistence checks this project ran extensively on the US mirror's 47-year history — there is no "pre-publication vs. post-publication" split possible here, and no way yet to check whether ASX momentum's edge is stable across sub-periods the way Sections 5.10-5.15 checked for the US. That is future work, not a result claimed here.

UPDATE FROM SECTION 5.24

Section 5.23 tested only momentum on ASX. Section 5.24 completes the picture: do 52-week-high and reversal replicate too?

5.24 Milestone 23 — Completing the ASX picture: do 52-week-high and reversal replicate too?

Section 5.23 tested only momentum on ASX. NSE and the US mirror were both tested on all three signals from the first pass (Section 5.4) — ASX had an incomplete picture by comparison. This milestone runs 52-week-high and short-term reversal on ASX with the identical out-of-sample hedge + HAC methodology, closing that gap.

	Long leg (daily / monthly)	Combined (daily / monthly)
52-week-high	+7.00%/yr, p=.099 / +8.41%/yr, p=.0091	+24.52%/yr, p=.0126 / +23.80%/yr, p=.0023
Short-term reversal	-4.04%/yr, p=.491 / -3.40%/yr, p=.529	-3.14%/yr, p=.818 / -2.49%/yr, p=.719

Reversal replicates the established pattern: no edge, anywhere. Consistent with Section 5.16's full retraction on NSE and the US mirror, reversal shows nothing on ASX either — not close to significant at either frequency, either leg. A third market, the identical null. This is a confirming result, not a new one, and needs no further qualification.

52-week-high does not — and this is a genuine update to a conclusion this project called "resolved" three signals ago. Sections 5.7-5.8 found 52-week-high's apparent edge on NSE and the US mirror was *entirely* a construction flaw: an uncontrolled long/short beta mismatch, with **zero** significant alpha in 12 of 12 hedged regressions across both markets and both frequencies. On ASX, the hedged combined book is significant at both frequencies (p=0.0126 daily, p=0.0023 monthly), and even the long leg alone clears significance monthly (p=0.0091). This is not the same failure mode Sections 5.7-5.8 found — the alpha survives the hedge here, rather than vanishing once beta is controlled for.

But this likely isn't a second, independent anomaly — it's the same mechanism as momentum's ASX result, viewed through a highly correlated signal. 52-week-high and 12-1 momentum are both trend-following constructions (recent winners vs. losers), and on ASX their leg returns are correlated at **0.76 (long leg) and 0.82 (combined book)** — they are substantially picking the same names. The short

leg's mean beta (-1.36) is close to momentum's own short-leg beta (-1.26) from Section 5.23, consistent with both signals' short sides being loaded with the same high-beta mining and resources stocks that underperformed through 2011-2015. The honest reading: ASX's 2011-2015 divergence was severe and persistent enough that *any* reasonable trend-following construction would have captured it, not that 52-week-high anchoring specifically is a real, independent behavioral edge on this market.

ONE MECHANISM, TWO CONSTRUCTIONS

Momentum remains this project's one demonstrably robust, independently-replicated finding across all three markets. 52-week-high's ASX result should be read as an open, mechanism-ambiguous finding — real and hedge-robust, but not yet shown to be *more* than momentum wearing a different construction — not folded in alongside momentum as a second confirmed ASX edge. The natural next check (momentum as an explicit control in the same regression) has not yet been run.

Part VI — From Research to Practice

6.1 An investment framework: the Behavioral Mispricing Score

The composite score from Part III.4 is designed to be used as a **tilt or flag layered on top of an existing fundamentals-driven screen**, never as a standalone strategy — the same pattern every fund in Part IV.5 actually follows (they all pair behavioral signals with fundamental discipline, not raw signal-following). A stock that screens well fundamentally *and* sits in the top behavioral decile is a stronger candidate than either signal alone would suggest; a stock flagged as a behavioral "loser" deserves a second look even when its fundamentals look fine.

DECISION & RATIONALE

Rule, stated as a direct consequence of Part V.3's finding, not as generic best practice borrowed from a textbook: **never trust a blended score without checking what each of its components does on its own**. The composite here looked mediocre; decomposing it found one component actively losing money with a 97% drawdown and another that at first looked like a genuine, positive, cost-adjusted edge (a claim Part V.9 later retracted). Any score this project's code produces should be reported and validated component-by-component before a blended version of it is trusted for a decision.

Decompose the legs before discarding a signal — still correct, but the fix isn't "long-only" after all. The long leg's raw average return is significantly positive, the short leg's significantly negative, in every specification tested (Part V.6). That decomposition rule stands. What it was used to conclude did not: Part V.7 found both legs' significant raw returns are fully explained by uncontrolled market-beta exposure (long $\approx +0.8$ beta, short ≈ -1.2 to -1.4 beta) in markets that rose $\sim 20\%$ /year over the sample — once beta is controlled for, **alpha is insignificant in all 12 regressions tested, both legs, both markets, both frequencies**. There is currently no demonstrated stock-selection skill in this signal, in either direction. **Corrected rule:** this signal is not a demonstrated source of alpha as currently built; using it as a behavioral tilt requires beta-neutralizing both legs first and re-testing the hedged residual for alpha — done in Part V.8, which independently confirmed the same conclusion with an actual rolling, out-of-sample hedge rather than a single regression coefficient. And more generally: check for a beta mismatch between a long-short book's legs before reaching for a behavioral explanation, not after three rounds of increasingly exotic ones.

UPDATE FROM PART V.9

Part V.9 applied the same beta check to the other two signals, and found something important in both directions. Short-term reversal — this project's one previously-reported positive finding — failed every one of its 12 alpha tests once beta was controlled for. It had never been re-checked with the standard of rigor developed for the losing signal; once it finally was, it didn't hold up either. **Rule: a positive finding earns no exemption from the checks a negative one gets.** But 12-1 momentum, tested the same way, showed something new: on the US mirror, its long leg and combined book carry large, highly significant alpha (annualized roughly +8-15%/yr, $p < 0.01$ in every cut), with the combined book's beta close to zero. This is currently the project's single most credible candidate for a genuine, demonstrated edge — not replicated on NSE, but a coherent, high-significance cluster rather than the isolated marginal hits this project has learned to distrust.

UPDATE FROM PART V.10

Part V.10 checked the one thing Part V.9 left open: publication decay. Real, textbook-consistent decay was found — alpha down 30-47% post-1994, one cut (combined book, daily) losing significance entirely — but not disappearance: three of four cuts remain significant through 2017 alone. **Size any claim about this finding on the decayed, post-1994 number, not the full-sample one.**

UPDATE FROM PART V.11

Part V.11 re-tested Part V.10's post-1994 number with an actual rolling, out-of-sample hedge (Part V.8's methodology) instead of an in-sample regression, and found it does not survive: post-1994 alpha is not statistically significant in either leg, at either frequency, once beta is estimated only from trailing data and re-hedged monthly. Pre-1994 alpha remains strongly confirmed. Treat this project's momentum finding as real and robust pre-1994, and not currently demonstrated to be forward-sizeable in the post-publication era.

UPDATE FROM PART V.12

Part V.12 dug into that null result at explicit request rather than stopping here: is it really just the 2009 momentum crash, or the hedge's rolling window? Neither explained it — a five-era breakdown showed ongoing decay through 2017, well after 2009, and the result held across three different hedge windows. This is the most rigorous read of momentum specifically: genuine, strong pre-1994 alpha, and genuine, ongoing decay since — not a temporary shock the strategy is due to recover from.

UPDATE FROM PART V.13

Part V.13 then asked whether this whole pattern is unique to momentum. At the time, it looked like it wasn't: short-term reversal's long leg showed the identical "real pre-1994, decayed since" signature ($p=0.046$ pre-1994, $p=0.629$ post), invisible inside Part V.9's full-sample null. 52-week-high, by contrast, showed no significant alpha in either era — it never had genuine skill at all. (*Part V.16 later retracted reversal's side of this finding entirely — see below.*)

UPDATE FROM PART V.14

Part V.14 then quantified the decay rate directly instead of trusting the 1994 cutoff, and found the two signals decay in genuinely different shapes — at the time. Reversal's decline looked like a real, smooth, statistically significant linear trend (zero-crossing ~November 2004). Momentum's was not smooth at all: alpha stayed flat and strong through 2008, then broke sharply, better described as a 2008-09 regime shift than gradual 1994-onward decay. *(Reversal's side of this was later retracted; momentum's was not — see Part V.16.)*

UPDATE FROM PART V.15

Part V.15 then formalized the 2008-09 claim with an actual structural-break test rather than a descriptive comparison. A single, literature-motivated Chow test at 2008-09-01 does confirm a real break in momentum ($p=0.026$) — but an unconstrained search corrected for multiple testing finds its best-fitting break 27 months later (Dec 2010) and that maximum is not significant (bootstrap $p=0.138$). Reversal appeared to show the opposite: nothing at 2008, but a genuine, significant break (bootstrap $p=0.048$) around **August 1980**. *(Part V.16 investigated that August 1980 date directly and found it was not a real market event — see below.)*

UPDATE FROM PART V.16

Part V.16 investigated the mechanism behind that August 1980 date at explicit request, rather than reporting it and moving on. It found reversal's decile long leg was only 2-4 stocks from 1972-83, drawn from a tiny, survivorship-biased universe, plus two previously undocumented data anomalies (WMT, INTC). **Reversal's entire positive-alpha claim — carried since Part V.10 — evaporates completely ($p=0.54-0.96$) once the unreliable 1972-1977 window is excluded, including from the exact date this section's own break search identified.** Momentum, tested as a control on the identical data, was unaffected ($p \leq 0.02$ throughout). Treat every reversal claim in this Part VI as retracted; momentum's claims stand.

UPDATE FROM PART V.17

Part V.17 then gave momentum's 2008-09 break a real mechanism: reapplying this project's own Bear $\times$ High-Vol momentum-crash regression (Part V.7-5.8, rejected for the 52-week-high signal) to momentum's hedged long leg, split at 2008-09-01. The interaction term is insignificant pre-2008 ($p=0.42$) but large, negative, and significant post-2008 ($p=0.008$): on Bear+High-Vol days, the long leg loses at roughly a 37% annualized rate post-2008, versus an insignificant ~11% pre-2008. **The classic momentum-crash mechanism was dormant before 2008 and activated after — a genuine causal story, not just a confirmed date.** *(Part V.19 later refined — did not retract — the "activated after" framing: see below.)*

UPDATE FROM PART V.18

Part V.18 then tested the same crash mechanism on NSE, and in the process exposed a gap in this project's own earlier NSE test. Part V.9's "no significant NSE momentum alpha" used a static regression, never the rolling out-of-sample hedge used for every US test since Part V.8. Applying that hedge to NSE for the first time reaffirms no significant full-sample alpha, but finds a tentative, "promising not confirmed" post-2008 signal (daily $p=0.044$) the cruder test missed. The Bear $\times$ High-Vol crash mechanism itself, however, does **not** replicate on NSE ($p=0.96$) — NSE shows a plain Bear effect instead, not the specific crash-rebound dynamic found for the US. *(Part V.22 later found this tentative NSE signal does not survive its own named universe-growth caveat — see below.)*

UPDATE FROM PART V.19

Part V.19 then split Part V.17's "post-2008" block at 2009-12-31 and found the trailing-12-month market return never went negative again after September 2009 through the end of the US sample — so the interaction term's pooled significance was carried entirely by the 337-day crisis window, and cannot even be tested against the ~2,000 days since (zero Bear+High-Vol observations there). Worse for the "interaction" framing specifically: run on the crisis window alone, the multiplicative term itself is not significant ($p=0.26$) — high volatility and bear-state each mattered independently instead. **The 2008-09 break is real; "activated since 2008" is not demonstrated — it is confirmed for the one crisis this sample's post-2008 window actually contains, and untested since, because no comparable bear market has recurred.** *(Part V.20 found this "no comparable bear market" claim itself depends on the 252-day lookback used to define "bear market" — see below.)*

UPDATE FROM PART V.20

Part V.20 swept the two parameters every regime-based finding since Part V.6 has used — the volatility window and the bear-market lookback — across a 4 $\times$ 4 grid. Part V.17's pattern (dormant pre-2008, active post) replicated in 8 of 16 combinations, reliably near this project's own defaults but not at the shortest windows tested. Part V.19's stronger claim — that no bear market recurred after 2009 — did **not** survive shorter, equally standard bear-lookbacks: at 126 or 189 trading days instead of 252, a bear regime fires post-2010, clustered in 2010-11 and 2015-16. Two different robustness verdicts for two different claims built on the same code. *(Part V.21 then ran the persistence test this made possible — see below.)*

UPDATE FROM PART V.21

Part V.21 re-ran the crash-mechanism interaction on the 2010-onward window under both alternate bear-lookbacks Part V.20 identified. It did not reactivate at either ($p=0.20$, $p=0.44$), and at the 189-day lookback the bear-market main effect was significant but **positive** — the opposite sign from crash risk. This strengthens, not just leaves open, Part V.19's "confirmed for one crisis" framing: the mechanism is now tested and not found outside 2008-09, in the two next bear-adjacent episodes this sample contains, not merely untested.

UPDATE FROM PART V.22

Part V.22 turned the pooled-window lesson from Parts V.19-21 back on this project's own remaining open finding: Part V.18's tentative NSE signal. Splitting at the date NSE's universe became permanently fixed at 48 names, neither the growing sub-period nor the fully stable sub-period is significant alone ($p=0.11$, $p=0.18$) — only the pooled full window is. Not the growth confound named, but the same pooling problem diagnosed in the crash mechanism, applied to this project's own house-favored open thread.

UPDATE FROM PART V.23

Part V.23 found and validated a genuinely independent third market (ASX Australia) after ruling out every reachable European source, and applied this project's current best methodology — the out-of-sample hedge, not the crude full-sample regression this project itself started with — from the very first test. Momentum's long leg came back significant at both daily ($p=0.0085$) and monthly ($p=0.0141$) frequency, the cleanest result of the three markets tested, on this project's shortest sample (six years).

UPDATE FROM PART V.24

Part V.24 completed the ASX picture. Reversal replicated its established null cleanly. 52-week-high did not: significant hedged alpha at both frequencies, reopening a question Parts V.7-8 had called resolved. But its ASX leg returns correlate 0.76-0.82 with momentum's own, and its short-leg beta (-1.36) nearly matches momentum's (-1.26) — the two signals are substantially picking the same names. Read as one mechanism (ASX's 2011-2015 mining divergence) viewed through two correlated constructions, not two independent confirmed anomalies.

6.2 A risk-management playbook, from the Q1 simulation

- **Never calibrate tail risk on a calm-regime correlation matrix alone.** The Part III simulation shows a Gaussian VaR model calibrated on "normal" data understates the true 99.9% tail loss by roughly 1.3x, purely because it cannot see correlations rising toward 1 under stress. Any leveraged strategy needs a stress-regime correlation scenario as a second, explicit check, not just a historical-calibration number.
- **Leverage does not just scale losses — it changes which losses are survivable.** LTCM's underlying spread moves were not physically unprecedented; 25:1+ leverage turned a bad quarter into insolvency. Position and leverage limits should be set against the stress-regime tail estimate, not the calm-regime one.
- **Crowding is a correlation risk invisible from inside your own portfolio.** The 2007 Quant Quake shows a well-diversified-looking book can be secretly correlated with every other fund running a similar signal. A practical mitigant: track how crowded a factor appears to be (e.g., the realized correlation of a strategy's returns to a public factor index) and de-risk when it's high, independent of the strategy's own apparent confidence.

- **A winning streak is not evidence against tail risk.** Amaranth's Brian Hunter had a strong record before the 2006 blowup; recent success under one regime is weak evidence a concentrated position is safe under a different one. Concentration limits need an explicit override that doesn't relax just because a book has been working.
- **Treat "the model says it's fine" as a hypothesis, not a fact** — especially near known regime-change triggers (sovereign defaults, liquidity crunches, crowded-factor unwinds). This is the project's central thesis applied reflexively to its own tooling, not just to the markets it studies.
- **The Q1 mechanism and the Q2 investigation asked the same question — Q1 answered it with a simulation, Q2's real data gave a more honest, weaker answer.** Part III's simulation showed, stylized, that tail risk compounds when volatility and correlation rise together. Part V.5's momentum-crash-risk investigation set out to find the same signature in a real backtest, and a first descriptive pass looked like it had. Formal significance testing (Part V.6) did not confirm it. Any short position built on a behavioral signal should still be regime-tested before being sized — that precaution doesn't depend on this specific mechanism being confirmed — but "regime-tested" has to mean the HAC-regression version, not the regime-bucket version, given what happened here when the two disagreed.
- **A compelling descriptive pattern is a hypothesis, not a finding — this project produced its own cautionary tale.** Part V.5's regime-bucket comparison looked like a clean, four-signature confirmation of momentum-crash risk. Formal HAC-regression testing at daily and monthly frequency (Part V.6) found none of the regime-conditioning coefficients significant in the leg the theory actually predicts, in either market. Both analyses used the same underlying data; only the statistical rigor differed. Never size a position, write a risk limit, or make a claim based on a descriptive regime split alone — run the regression with proper standard errors first, and expect a real chance the compelling-looking pattern won't survive it.
- **The cheapest test is the one to run first, and this project ran it last.** Three escalating hypotheses (crash-window concentration, a value/growth confound, formal momentum-crash regime-conditioning) were tested across three milestones before Part V.7 finally ran a plain CAPM beta regression — the single cheapest, most standard check for any long-short book — and found the entire answer in it: an uncontrolled beta mismatch between the legs, fully explaining every prior milestone's numbers, no behavioral story needed. Order hypothesis tests from cheapest and most mechanical to most exotic, not the reverse; a beta regression takes minutes and rules out the most common cause of surprising long-short performance before any more elaborate explanation is worth reaching for.
- **A risk process that only re-tests its losers eventually trusts a winner it never should have.** Every rigor upgrade through Part V.8 was applied to the signal that was losing money. The one signal reported as a genuine edge (reversal) rode on its original, less rigorous validation for six milestones before anyone checked it the same way. Part V.9 finally did, and it didn't hold up. Schedule periodic re-validation of every "confirmed" edge under your *current* standard of rigor, not just the standard at the time it was confirmed — a standard that improves over a book's life should apply retroactively, especially to positions still being sized on the old conclusion.
- **Size a position on the decayed number, not the full-sample number — and check that decayed number was itself estimated out-of-sample.** Part V.10 split the one surviving edge (US

momentum) at its 1993 publication date and found real decay: alpha down 30-47% post-1994, one cut losing significance outright. That was already a large correction — but it was still an in-sample estimate. Part V.11 re-tested the same period with an actual rolling, out-of-sample hedge and found the post-1994 alpha does not survive at all, in either leg. The full-sample average was never the honest number to size against once a signal is old enough to have a publication date, and an in-sample per-era regression isn't the final word either — only a hedge that could actually have been traded forward tells you what capital entering today can expect.

- **Before accepting a null result, rule out the obvious "maybe it's just one bad episode" and "maybe it's a methodology artifact" objections — and if it survives both, treat that as the null result getting stronger, not weaker.** Part V.12 tested whether Part V.11's post-1994 null result was really just the 2009 momentum crash, and whether it depended on the specific hedge window chosen. Neither objection explained it: the era-by-era trend showed ongoing decay through 2017, well after 2009, and the result was identical across three different hedge windows. A finding that survives a genuine attempt to explain it away deserves more confidence, not less — the temptation after an unwelcome result is to look for the one adjustment that makes it go away; running that check honestly, and reporting it even when it doesn't help, is what separates a stress-test from a fishing expedition.

- **A full-sample "no significant alpha" verdict does not mean a signal was never real — check whether it's actually two eras averaging to zero before writing it off entirely.** Part V.9 declared short-term reversal fully retracted based on a full-sample regression, the technically correct read of that specific test. Part V.13 applied the era-split toolkit built for momentum and found reversal's long leg had been genuinely, significantly positive pre-1994 and decayed to noise since — the same pattern as momentum, hidden inside a full-sample average that happened to net out near zero. Retest every full-sample null result from before your era-split toolkit existed with that toolkit, not just your full-sample findings — a "no effect on average" verdict can quietly contain a real, decayed effect that a single-number summary cannot distinguish from a signal that was simply never real. *(This specific reversal example was itself later retracted by Part V.16 — see below — but the underlying rule, that a full-sample null deserves a second look with an era-split toolkit, still stands.)*

- **A binary before/after split can get the direction right while getting the mechanism wrong — quantify the trend continuously before naming a cause.** Part V.14 fit a continuous linear decay rate to momentum's and reversal's hedged long legs instead of trusting the 1994 publication-date cutoff. Reversal's decay appeared, at the time, to be genuinely smooth and statistically significant, consistent with the publication-decay story — later shown by Part V.16 to be an artifact, not a real trend. Momentum's trend was not significant, because the real pattern is a sharp break around the 2008-09 financial crisis, not a gradual erosion starting in 1994 — and that finding held up. Once a binary split finds a real effect, don't stop there — fit a continuous trend and inspect a rolling, non-parametric trajectory to check whether the story you're about to tell actually matches the shape of the data, or just happens to fall on the correct side of an arbitrary cutoff.

- **Even a data-driven, non-arbitrary date can still be the wrong test — correct for the search itself before trusting it.** Part V.14 picked September 2008 by eyeballing a rolling trajectory, which is a comparison, not a test: it doesn't say whether that split is meaningfully better than what chance alone would produce from searching many candidate dates. Part V.15 ran the actual test two ways: a single pre-registered date (motivated by an external, published crash episode, not this project's own plot) confirmed momentum's break; an unconstrained search over ~375 candidate dates, corrected

via bootstrap for having searched that many, did not decisively confirm any single dominant break (its own best-fit date, December 2010, wasn't even the one the earlier milestone had proposed). A hypothesis motivated by an external, independent source can be tested directly and cheaply; a hypothesis motivated by your own data requires a multiple-testing-corrected test before it earns the same confidence — and expect the corrected version to be measurably more conservative.

- **When a signal's finding depends on data density you haven't checked, check it before trusting the finding.** Part V.16 traced reversal's August 1980 "break" to its source: a decile portfolio of only 2-4 stocks, drawn from a 9-14-name universe of today's mega-cap survivors backfilled to the 1970s — plus two previously undocumented data anomalies (an apparent unadjusted-split round-trip in WMT, two large jumps in INTC). Reversal's entire positive-alpha claim depended on this unreliable window and vanished completely once it was excluded, at every later start date tested. Momentum, run through the identical thin data as a control, was unaffected. A decile portfolio with 2-4 names is not a diversified strategy, it is a handful of individual stock bets, and "statistically significant" on such a sample tells you about those specific stocks' survivorship, not about a market-wide behavioral effect. Check minimum portfolio size and data density for every sub-period a significance claim rests on, not just the full sample's average.

- **A structural break confirmed by a date is still an unexplained fact until tested against a named mechanism.** Part V.17 applied this project's own Bear $\times$ High-Volatility momentum-crash regression (Parts V.7-8, rejected for the 52-week-high signal, full-sample) to momentum's hedged long leg, split at the same 2008-09-01 date. The interaction term is insignificant pre-2008 ($p=0.42$) but large, negative, and significant post-2008 ($p=0.008$) — the classic momentum-crash mechanism was dormant through the pre-crisis decades and activated since. A mechanism this project rejected for one signal, in one era, can still be real for a different signal in a different era — don't let an earlier rejection close off re-testing the same hypothesis somewhere new; a confirmed break date is a fact, not yet an explanation.

- **"Already checked" and "checked with your current best method" are not the same claim.** Part V.9's NSE momentum test used a static full-sample regression. Applying the rolling out-of-sample hedge this project built in Part V.8 to NSE momentum for the first time reaffirmed no significant full-sample alpha, but surfaced a tentative post-2008 signal (daily $p=0.044$, monthly $p=0.073$) the cruder test lacked the power to see — while the Bear $\times$ High-Volatility crash mechanism confirmed for US momentum (Part V.17) turned out not to replicate on NSE at all. A rejected finding on an already-tested market is worth re-checking with each new methodology upgrade this project builds, not just applying new methods to new markets — the earlier rejection may have been correct for the test it used and still be missing something a better test would find; and a confirmed mechanism on one market is a hypothesis, not a law, everywhere else.

- **A pooled multi-year regression's significant coefficient can be one crisis acting as its own control group, not a persisting regime.** Part V.19 split Part V.17's "post-2008" test window at end-2009 and found the US mirror's trailing-12-month market return never went negative again after September 2009 through the end of the sample — so the interaction term found significant across the whole post-2008 block had zero Bear+High-Vol observations to draw on outside the 337-day crisis itself. Run on the crisis window alone, the multiplicative interaction specifically was not significant ($p=0.26$); volatility and bear-state each mattered on their own instead. Before reading a coefficient significant across a multi-year window as evidence of a standing regime, split the window and check

whether the conditioning regime actually recurs throughout it — a test that silently rests on one sub-period contrasted against a calm remainder is a test of "did something happen once," not "is this now a permanent feature."

- **A qualitative pattern and the specific quantitative claim built on top of it can have very different robustness, and a binary "did X ever happen" claim is only as robust as its own definition.** Part V.20 swept the two regime-construction parameters every crash-regime finding since Part V.6 has used — a 4×4 grid of volatility windows and bear-market lookbacks. Part V.17's "dormant pre-2008, active post-2008" pattern replicated in 8 of 16 nearby parameter combinations, reliably near this project's own defaults. But Part V.19's specific "no bear market recurred after 2009" claim did not survive shorter, equally standard bear-lookbacks (126 or 189 trading days instead of 252) — a bear regime fires post-2010 under those conventions. Test the pattern and the specific claim separately, since they can have different robustness; and when a finding rests on whether some condition ever occurred, stress-test the condition's own definition, not just the finding built on top of it — the definition is often the more arbitrary choice.

- **"Untested" and "tested and not found" are different claims, and closing a robustness thread means running the test it made possible, not just noting that it's now possible.** Part V.21 re-ran the crash-mechanism persistence test under the two bear-lookbacks Part V.20 found actually see a post-2010 bear regime. The interaction did not reactivate at either window ($p=0.20$, $p=0.44$), and at the 189-day lookback the bear-market main effect was significant but positive — the opposite sign from crash risk. When a robustness check reveals that a previously untestable question has become testable, run the test rather than leaving it flagged as an open thread — a mechanism that fails to reactivate when given a real chance to is materially stronger evidence than a mechanism that was simply never checked.

- **A named caveat is a promise to test it, and a lesson learned from one correction applies to your own other findings, not just the one that taught it.** Part V.22 tested the growth-confound caveat Part V.18 attached to the tentative NSE momentum signal, splitting at the exact date NSE's universe became permanently fixed at 48 names. The growing sub-period showed a larger point estimate, not a smaller, thin-universe-inflated one, so the named confound wasn't confirmed — but neither sub-period reached significance alone, only the pooled full window, the identical pooled-window pattern Parts V.19-21 had just diagnosed in the crash mechanism. Apply a methodology lesson learned from one finding to every other finding built the same way, including your own project's most-favored remaining result — a named but untested caveat is unfinished work, not a disclosed limitation.

- **A sandbox's specific network allowlist is itself a research constraint worth documenting, and a genuinely independent third market is worth the search effort even when the first choice isn't reachable.** Part V.23 spent real effort ruling out Stooq, Hugging Face, *github.com*'s own pages, and general *api.github.com* browsing before finding *raw.githubusercontent.com* could serve a real, committed ASX dataset once the exact file paths were known. Momentum replicated cleanly there ($p=0.0085$ daily, $p=0.0141$ monthly), the strongest of the three markets, on a sample too short (six years) for the era-stability checks run on the US mirror. Document a data search's dead ends, not just its destination — the next milestone benefits from knowing which doors were tried and found locked; and label a new result by what it has and hasn't yet been checked against, not by how clean it looks on first pass.

- **A correlated pair of signals confirming together is not two confirmations — check whether a new result is independent of an already-established one before counting it separately.** Part V.24 completed the ASX picture: reversal replicated its established null cleanly, but 52-week-high showed significant hedged alpha where Parts V.7-8 had found zero alpha in 12 of 12 regressions on the other two markets. Checking the correlation between 52-week-high's and momentum's ASX leg returns (0.76 long leg, 0.82 combined) and their near-identical short-leg betas (-1.36 vs. -1.26) showed the two signals were substantially picking the same names. Before treating a new significant result as a second, independent finding, check its correlation with an already-confirmed one built from related inputs — a shared underlying driver can make one real economic effect look like two separate discoveries.

6.3 A standalone business idea: decomposed behavioral signal analytics

Off-the-shelf factor data (momentum, value, quality) is typically sold by large vendors as pre-blended, black-box composites, priced for institutions with large budgets. Smaller systematic funds, family offices, independent RIAs, and research desks either cannot afford that tier, or cannot see *inside* the composite to know which component is actually carrying the edge on their specific universe — exactly the failure mode this project hit directly on the NSE data (Part V.3), and hit again when its own first-reported positive finding (reversal) failed a beta check the project hadn't yet thought to run (Part V.9). Decomposed, beta-adjusted validation is what surfaced the project's one genuine finding (US momentum) instead of its retracted one.

DECISION & RATIONALE

This idea was originally framed as an extension of the specific software platform this research happened to be built alongside. That framing was dropped after explicit feedback that the business idea should stand on its own, independent of any particular existing product. The pitch below reflects that: a standalone service, differentiated on the same transparency principle (decomposed, auditable signals; never a pre-blended black box) that Part V.3's finding demonstrated the value of directly.

The product: a subscription analytics service with two parts — a **signal side** (the momentum / 52-week-high / reversal library, run per client against *their* universe, reported as separate, individually-backtested components, never pre-blended, each with its own beta-regression alpha/beta breakdown so a client can see whether performance is genuine stock-selection skill or just uncontrolled market exposure — Part V.7's finding made this check a required feature, not an optional one — and, for any signal drawn from published academic research, a pre/post-publication decay split re-confirmed with an actual out-of-sample hedge, not just an in-sample regression (Parts V.10-11's check), and a minimum-data-density flag on any period where a decile portfolio would hold fewer than roughly 10 names (Part V.16, which found a "genuine" edge that was actually 2-4 survivor-biased stocks) — differentiators most factor-data vendors don't surface at all) and a **risk side** (the regime-switching stress-VaR methodology from Part III, run against a client's actual position correlations and leverage, reporting calm-regime vs. stress-regime tail loss side by side).

Target customer: small-to-mid systematic equity funds, family offices, and independent RIAs priced out of institutional factor-data tiers but sophisticated enough to want decomposed, re-validated signals rather than a black box; secondarily, finance graduate programs and CFA/PE prep courses, as a teaching tool for the exact "don't trust the blend" lesson this project surfaced.

Revenue model: a per-seat analytics subscription, tiered by number of tracked universes/portfolios — the same go-to-market as existing quant factor-data vendors, priced and scoped for the segment those vendors serve poorly, differentiated specifically on transparency: every number traces to runnable code and a stated backtest window, not a proprietary methodology.

Moat: not the signals themselves (all public, published research) — the moat is the discipline of per-client, per-universe decomposed validation instead of a generic pre-blended score, which is exactly what a larger vendor selling a standardized product across all clients structurally cannot do cheaply.

Part VII — Limitations and Honesty Statement

Stated together in one place because a project whose entire thesis is "don't trust a model's output just because it looks rigorous" owes the same discipline to its own outputs.

- **Survivorship bias.** Both the NSE mirror and the US mirror located in milestone 3 use a present-day-chosen company list, not point-in-time index membership. A rigorous version needs a survivorship-bias-free universe (e.g., CRSP, or a maintained point-in-time constituents file).
- **Data provenance.** Both the NSE and US mirrors are personal, community-uploaded GitHub repositories, not official exchange or licensed vendor feeds. Results from them are a genuine methodology demonstration on real prices, not investment-grade research, until cross-checked against an official source.
- **Momentum and reversal did not replicate consistently** across the two markets tested (Part V.4) — treat either result, in isolation, as provisional rather than a confirmed anomaly. Only the 52-week-high signal's loss held up in both.
- **Short-term reversal's positive result is retracted, fully, at every level this project checked (Parts V.9, then V.13, then V.16 — a claim that briefly looked rehabilitated before being retracted again, more thoroughly).** Part V.9's full-sample CAPM beta check found none of reversal's 12 alpha tests significant. Part V.13 then found this incomplete: reversal's long leg carried a real-looking, significant pre-1994 alpha ($p=0.046$) hidden inside that full-sample average. But Part V.16 traced that apparent pre-1994 alpha to its source and found it depends entirely on a severely thin (2-4 stock), survivorship-biased, partly data-glitched 1972-1977 sample window — it vanishes completely ($p=0.54-0.96$, often slightly negative) from any start date at or after 1978, including the exact date Part V.15's own structural-break search identified as reversal's "true" break (1980-08-29, $p=0.922$). Momentum, tested through the identical unreliable early data as a control, was unaffected ($p \leq 0.02$ at every comparable start date). **Reversal shows no reliably demonstrated edge anywhere in this dataset, in any leg, at any point in the sample, once properly checked.**
- **US 12-1 momentum's post-1994 alpha does not survive an actual out-of-sample hedge, and the deeper investigation confirms genuine decay rather than a crash or methodology artifact (Parts V.10-12, superseding an earlier, more optimistic in-sample reading).** An in-sample pre/post-1994 split (Part V.10) found real decay but apparent survival in three of four cuts. Re-testing the same post-1994 period with a rolling, out-of-sample beta hedge found no statistically significant alpha in either leg, at either frequency (Part V.11); the hedged combined book's post-1994 average return is outright negative. Part V.12 then tested whether that null result was just the 2009 momentum crash or a hedge-window artifact: an era-by-era breakdown showed ongoing decay through 2017 (well past 2009), excluding the crash window left the combined book's result essentially unchanged, and the null result held across three different hedge windows. Pre-1994 alpha remains real and robust under every test. This project's momentum finding should be read as strong and genuine before 1994, and as genuine, ongoing decay — not a temporary shock — since.
- **"Real pre-1994, decayed since" is not momentum-specific (Part V.13).** Applying the identical pre/post-1994 out-of-sample hedge to all three US signals found two distinct stories: 52-week-high shows no significant alpha in either era, in any leg (it never had genuine stock-selection skill);

reversal's long leg shows the identical decay signature as momentum's. The pattern appears in two of three signals' long legs, not one, consistent with a market-wide explanation rather than a momentum idiosyncrasy. *(Since retracted for reversal — Part V.16 traced this "pattern" to a thin, survivorship-biased early sample; momentum's finding is unaffected.)*

- **The 1994 cutoff was directionally right for momentum but misdescribes its shape and timing (Part V.14).** A continuous linear-trend regression on momentum's hedged long leg gives a slope that is not statistically significant ($p=0.15$) — the effect did not decay smoothly. A rolling 5-year trajectory shows momentum's alpha stayed consistently strong (+4% to +22%/yr) from the 1970s through the window ending January 2008, then broke sharply negative from 2009 on. Splitting at September 2008 gives a far cleaner divide (pre: +8.02%/yr, $p=0.0005$; post: -0.56%/yr, $p=0.998$) than 1994 ever produced. Reversal's decay, by contrast, is well-described by a smooth linear trend (slope -0.41%/yr, $p=0.026$, zero-crossing ~November 2004). Momentum's weakness is better attributed to a 2008-09 regime shift than to slow, 1990s publication-driven crowding. *(Since retracted for reversal — Part V.16 found this "smooth trend" was itself an artifact of unreliable early data; momentum's finding is unaffected.)*

- **A formal structural-break test qualifies both of Part V.14's stories further (Part V.15).** A Chow-style test at the single, literature-motivated date (2008-09-01, from Daniel & Moskowitz 2016) confirms a real level shift in momentum ($p=0.026$ daily) — but an unconstrained Quandt-Andrews sup-Wald search (which does not assume any break date) finds its single best-fitting break 27 months later, at December 2010, and that unconstrained maximum is not statistically significant once corrected for the multiple-testing problem of searching ~375 candidate dates (400-draw block-bootstrap $p=0.138$). Reversal shows the opposite: no break near 2008 ($p=0.34$), but a genuine, significant break (bootstrap $p=0.048$) around **August 1980** — the sharp early decline from reversal's extraordinarily high late-1970s level, not a smoothly accumulating multi-decade slope as the linear-trend regression alone suggested. Treat both signals' "when did it break" story as more conservative and more precisely qualified than Part V.14's descriptive framing. *(Since retracted for reversal — Part V.16 found this August 1980 "break" is a thin-universe and data-quality artifact, not a real 1980 market event; momentum's finding is unaffected.)*

- **What actually happened in 1980? Not a market event — a thin, survivorship-biased universe plus a data-quality defect (Part V.16).** Investigating the mechanism behind Part V.15's August 1980 reversal break at explicit request, rather than reporting the date and moving on, found reversal's decile long leg was only 2-4 stocks from 1972-1983, drawn from a 9-14-name universe of today's mega-cap survivors backfilled to their earliest available data, plus two previously undocumented data anomalies (WMT round-trips -52%/+109% across two weeks in December 1974; INTC jumps +101% and +51% in 1972, both consistent with split-adjustment errors). Re-testing reversal's long-leg significance from a range of start dates shows its entire positive-alpha claim (already only marginal at full sample, $p=0.068$) depends completely on the unreliable 1972-1977 window: from any later start, including the exact date Part V.15's own search identified, alpha is never significant ($p=0.54$ to 0.96, usually slightly negative). Momentum, run through the identical thin, biased, partly-glitched early data as a control, was unaffected ($p\leq 0.02$ at every comparable start date). **Short-term reversal's apparent pre-2005 alpha is retracted outright, not narrowed: this project has found no reliably demonstrated reversal edge anywhere in this dataset, in any leg, at any point in the sample.**

- **Momentum's 2008-09 break now has a concrete causal mechanism, not just a confirmed date (Part V.17).** Applying this project's own look-ahead-free Bear × High-Volatility momentum-crash regression (Parts V.7-8, where it was rejected for the 52-week-high signal, full-sample) to momentum's hedged long leg, split at 2008-09-01: the interaction term is small and insignificant pre-2008 ($p=0.42$) but large, negative, and significant post-2008 ($\text{coef}=-0.00218/\text{day}$, $p=0.008$) — on the ~9% of post-2008 trading days that are both high-volatility and trailing-bear, the long leg loses at an annualized rate around 37%. The classic momentum-crash mechanism this project rejected for the original signal turns out to be real for momentum specifically, but conditional on era: dormant through the pre-crisis decades, active since. (*Refined, not retracted, by Part V.19: the "active since" framing overstates what was tested — see below.*)
- **The momentum-crash mechanism does not replicate on NSE — but a more rigorous re-test finds a tentative NSE momentum signal a cruder test had missed (Part V.18).** Part V.9's "no significant NSE momentum alpha" used a static full-sample regression; applying the project's own rolling out-of-sample hedge (used for every US momentum test since Part V.8) to NSE momentum for the first time reaffirms no significant alpha full-sample (daily $p=0.170$, monthly $p=0.168$), but surfaces a new, "promising, not confirmed" post-2008 signal (daily $p=0.044$, monthly $p=0.073$, marginal) — with the real caveat that NSE's universe itself grew from ~30 to 48 names over this window, so part of the improvement may reflect a less thin cross-section rather than a genuine regime change. (*Downgraded by Part V.22: testing this exact caveat found neither sub-period significant alone — see below.*) Separately, Part V.17's exact Bear × High-Vol interaction is never significant on NSE ($p=0.96$ full-sample) — the specific crash-rebound mechanism found for US momentum does not generalize. NSE does show a plain, unconditional Bear effect instead (loses significantly in any trailing bear market, $p=0.002$ full-sample) — a related but mechanistically different, more generic bear-market sensitivity.
- **Milestone 16's "post-2008 crash regime" is really "the one crisis in the post-2008 window," not a standing feature — and in isolation, the interaction term itself is not what's significant (Part V.19).** Splitting the post-2008-09 block at 2009-12-31: the US mirror's trailing-12-month market return never went negative again after September 2009, through the end of its coverage (Nov 2017) — the 2011 and 2015-16 selloffs both recovered before the 252-day lookback registered them as sustained bear states. That means Part V.17's "post-2008" test is entirely carried, for the interaction term, by the 337-day 2008-09 crisis window; the other ~2,000 post-crisis days contribute zero Bear+High-Vol observations, so the interaction cannot even be estimated there (rank-deficient regression). Worse for the "interaction" framing specifically: run on the crisis window alone, the multiplicative term itself is *not* significant ($p=0.26$ long leg, $p=0.31$ combined) — high volatility alone ($p<0.0001$) and the bear flag alone ($p=0.036$ - 0.005) do the work instead. Read Part V.17's finding as: real for the 2008-09 crisis specifically, evidence for volatility and bear-state each mattering independently more than for their interaction being the mechanism, and untested — not disconfirmed — as a standing post-2008 regime, since no comparable bear market has recurred in this sample to test against. (*Part V.20 found this "no comparable bear market" claim itself depends on the 252-day lookback used to define "bear market" — see below.*)
- **The crash-mechanism pattern replicates across most, but not all, nearby regime-construction choices — and the specific claim that no bear market recurred after 2009 does not (Part V.20).** Sweeping a 4×4 grid of volatility windows (10/21/42/63 days) and bear-market

lookbacks (126/189/252/378 days): Part V.17's pattern holds in 8 of 16 combinations, reliably (3 of 4 lookbacks) at this project's own default 21-day volatility window and the adjacent 42-day window, but never at the shortest 10-day window and rarely at the shortest 126-day bear-lookback. The default combination Parts V.17-19 actually used sits inside the reliable region, not at a cherry-picked edge case. But the bear-regime-frequency finding underlying Part V.19 is fragile: a bear regime fires post-2010 at both shorter lookbacks tested (126 and 189 days, clustered in 2010-11 and 2015-16) and never at the two longer ones (252, the default, and 378) — so whether the crash mechanism has a second data point to test persistence against depends on which standard convention is chosen, and that persistence test has not yet been re-run under the windows where a second bear regime actually exists. (*Closed by Part V.21: re-run under both alternate lookbacks, the mechanism did not reactivate — see below.*)

- **The crash mechanism did not reactivate in 2011 or 2015-16, even once real bear-regime days exist to test it against (Part V.21).** Re-running the Bear × High-Vol interaction on the 2010-onward window under the two lookbacks (126 and 189 days) where a bear regime actually recurs: the interaction is not significant at either window ($p=0.203$ at 126 days, $p=0.443$ at 189 days), despite 148 and 31 Bear+High-Vol days respectively to test it against. At the 189-day lookback the bear-market main effect *is* significant ($p=0.023$) but positive — momentum's long leg did somewhat better, not worse, during those trailing-bear periods, the opposite sign from both the 2008-09 crisis coefficient and what the crash-risk mechanism predicts. This strengthens, rather than merely leaves open, Part V.19's "confirmed for one crisis" framing: the mechanism is now tested and not found outside 2008-09, not just untested. Whether it would reactivate in a genuinely severe future crisis, as opposed to the milder 2011 and 2015-16 episodes, remains open — this sample has not contained one since 2009.

- **The tentative NSE post-2008 momentum signal (Part V.18) does not survive splitting by universe stability — its own named caveat, finally tested (Part V.22).** NSE's universe grew from ~30 names in 2000 to a fixed 48 by 2010-11-04, then stayed at exactly 48 names every day through the end of the sample. Splitting Part V.18's post-2008 window at that date: the growing sub-period (2008-09 to 2010-11, $n=535$ days) shows a *larger* point estimate (+23.72%/yr) than the full window but is not significant ($p=0.112$) on its own; the fully stable, fixed-48-name sub-period (2010-11 onward, $n=2,579$ days) shows a smaller point estimate (+4.60%/yr) and is also not significant ($p=0.180$). Neither direction of the growth-confound concern is confirmed — the growing years are not where the significance is concentrated — but a different, arguably more serious problem is: the full-window significance ($p=0.044$) exists only when the two sub-periods are pooled, echoing the exact pooled-window lesson Parts V.19-21 learned about the crash mechanism, applied here to this project's own remaining open finding. The NSE post-2008 signal should now be read as not currently demonstrated, not merely as an unconfirmed but promising thread.

- **Momentum replicates on a third, independent market (ASX Australia), the cleanest result of the three tested — but on this project's shortest history, and finding the market at all took real effort (Part V.23).** A thorough search for a European per-company OHLCV mirror found none reachable from this sandbox: *stooq.com*, *huggingface.co*, *github.com*'s own HTML pages, and general *api.github.com* repo browsing are all blocked, and several candidate repositories turned out to be fetch-at-runtime pipeline code (also blocked), not committed data. *grantcarthew/data-asx-historical-share-tables* — a mirror of ASX's own daily S&P/ASX300 report

emails, 2009-10-20 to 2015-12-31 — was the source that worked. Applying this project's out-of-sample hedge and HAC significance test directly (not the cruder full-sample regression Parts V.7-9 started with): long-leg alpha +11.14%/yr daily ($p=0.0085$) and +9.97%/yr monthly ($p=0.0141$); the combined book shows an even larger, still-significant effect ($p<0.001$ both frequencies), plausibly tied to Australia's 2011-2015 mining-sector downturn loading the short leg with high-beta losers (mean short-leg beta -1.26 vs. the long leg's +0.83) rather than being a hedge-instability artifact (rolling beta std 0.39, no extreme outliers). This is a real limitation, not just a caveat: six years is enough for a full-sample significance test but not for the era-splits, structural-break tests, or persistence checks Parts V.10-15 ran on the US mirror's 47-year history — ASX momentum has not yet been checked for stability across sub-periods the way every other surviving finding in this project has.

- **The 52-week-high result's cause: resolved, and it's the boring answer.** Two behavioral/statistical explanations were ruled out by direct test: crash-window concentration (Part V.4) and a value/growth confound (Part V.5), neither surviving in either market; formally-tested momentum-crash risk (Part V.6) then walked back Part V.5's descriptive-looking confirmation too. The actual explanation (Part V.7) is a construction flaw, not a market phenomenon: the legs have significantly different, uncontrolled market-beta exposure (long $\approx +0.8$, short ≈ -1.2 to -1.4), leaving the book net short-beta (-0.32 to -0.70 , $p<0.001$ every cut) in markets that returned $\sim 20\%$ /year. Once beta is controlled for, alpha is insignificant in all 12 regressions tested. There is no demonstrated stock-selection skill in this signal, in either direction, as currently built. (*Reopened, not overturned, by Part V.24: ASX shows significant hedged 52-week-high alpha, but it correlates 0.76-0.82 with momentum's own ASX result — likely the same mechanism, not new independent skill. See below.*)
- **52-week-high shows significant hedged alpha on ASX — reopening, not confirming, a question Parts V.7-8 called resolved — but it is highly correlated with momentum's own ASX result, not clearly a second independent edge (Part V.24).** On ASX, the hedged combined book is significant at both frequencies ($p=0.0126$ daily, $p=0.0023$ monthly) — the alpha survives the hedge this time, a different failure mode than before. But 52-week-high's and momentum's ASX leg returns correlate at 0.76 (long leg) and 0.82 (combined book), and the short leg's mean beta (-1.36) closely matches momentum's own (-1.26) — both signals are substantially picking the same names, consistent with both simply capturing Australia's 2011-2015 mining-sector divergence rather than 52-week-high anchoring being an independent behavioral edge on this market. Reversal, tested alongside it, shows nothing ($p=0.49$ - 0.82 across all four cuts) — a clean confirmation of its established null. Treat 52-week-high's ASX result as open and mechanism-ambiguous, not as a second confirmed ASX anomaly alongside momentum — the natural next check (momentum as an explicit control in the same regression) has not yet been run.
- **Transaction costs are a simple linear model**, not a real market-impact model; a strategy sized for real capital would need a proper implementation-shortfall estimate.
- **No out-of-sample / walk-forward validation is wired up by default.** A real deployment should split into a strict in-sample fit period and out-of-sample test period, and check for performance decay after each anomaly's academic publication date — a well-documented risk for momentum and reversal specifically.
- **The Q1 simulation's parameters are illustrative**, calibrated loosely to the LTCM episode's qualitative shape (correlations that were low/moderate in normal times moving toward 1 in crisis), not

fit to LTCM's actual, never fully disclosed book. Its reported multiple should be read as an order of magnitude, not a precise historical reconstruction.

- **The beta-hedge test (Part V.8) is itself imperfect, by design and by honest admission.** The rolling hedge leaves a small residual correlation with the market (-0.02 to -0.07, not exactly zero), the estimated beta is quite unstable across time in both markets (its standard deviation rivals its mean), and no hedging transaction costs are modeled. None of that changes the direction of the finding (still no significant residual alpha), but a live implementation would need a more robust hedging scheme and would bear real costs this analysis doesn't capture.

Conclusions

The Efficient Market Hypothesis is a useful default, not a law of nature. This project's five case studies show it failing in genuinely different ways — a Nobel-laureate-staffed fund undone by other investors' correlated panic; sophisticated quant funds undone by their own hidden crowding; an entire industry's risk model undone by a correlation assumption nobody's calibration data ever tested; a struggling retailer's stock undone (in the institutional short-sellers' favor, this time) by coordinated retail attention — which is itself evidence against any single, tidy story of "why markets are irrational." There isn't one mechanism; there are several, each traceable to a specific, well-documented piece of human psychology.

The project's real empirical result (Part V.3) delivered a finding more useful than a simple confirmation of either extreme view would have been: not "markets are irrational, biases always pay," and not "the anomalies are all arbitrated away, don't bother." Instead: one specific, named bias (anchoring, via the 52-week-high signal) actively lost money on this universe, while a different one (overreaction, via short-term reversal) at first looked like a real, cost-adjusted edge — and the two were invisible from inside a single blended composite. That is, in miniature, the entire project's argument: rigor means checking the specific mechanism, not trusting the reassuring-looking aggregate. It took until Part V.9 for the project to apply that same rigor to its own positive-looking half of the story, and when it did, the "edge" didn't survive either — but a different signal, US 12-1 momentum, did, and Part V.10 went one step further than any earlier milestone: instead of asking whether the finding was real, it asked how much of it a trader could still expect to capture today, and found a genuinely mixed, honest answer at first — real decay, not disappearance. Part V.11 pushed the same question one step further still, replacing an in-sample estimate with an actual out-of-sample hedge, and the honest answer got more sobering: no statistically significant post-1994 alpha survives in either leg. Even this project's own most reassuring recent finding kept shrinking every time it was checked with a sharper tool. Part V.12 then did something different from every check before it: instead of applying a sharper tool and finding the result shrink again, it tried directly to explain the shrinkage away — as a single crash episode, or as an artifact of one modeling choice — and failed on both counts. The result held. Part V.13 then went back to the signal Part V.9 had seemed to retract outright — short-term reversal — and applied the same era-split toolkit rather than trusting the earlier full-sample verdict, and found reversal's long leg had been telling the truth about being "a real, cost-adjusted edge" all along, just not for the era that mattered by the time anyone checked: genuine pre-1994, decayed since, the same shape as momentum's story — or so it seemed. Part V.14 finally asked the question every prior milestone had assumed the answer to without checking: was "the same shape" actually true? It wasn't. Quantified as a continuous trend instead of a binary 1994 split, reversal's decline turned out to be genuinely smooth, but momentum's did not — momentum held flat and strong for 35 years and then broke sharply around the 2008-09 crisis, a shape a decay curve fits badly and a regime shift fits well — or so a descriptive comparison suggested. Part V.15 finally asked whether that comparison would survive being turned into an actual test, corrected for the fact that the best of many candidate break dates always looks impressive. It survived only partly: the externally-motivated 2008-09 hypothesis for momentum held up, but an unconstrained search for the single best break in the whole series landed 27 months later and wasn't significant once that search was accounted for — and reversal's "smooth" decline turned out to hide a real, significant break of its own, decades earlier than anyone had looked. Part V.16 then did what none of the previous checks had: it asked *why* that August 1980 date, rather than reporting it as a fact and moving on. The answer was not a market event. It was a decade

portfolio of two to four stocks, drawn from a universe of today's mega-cap survivors backfilled to the 1970s, compounded by two previously undiscovered data anomalies. Reversal's entire positive-alpha claim — the one this project had been calling "genuine" since Part V.10, narrowing and re-narrowing it seven separate times without ever asking whether it was real — evaporated the moment the unreliable years were excluded, at every start date tested, including the exact date its own break search had identified. Momentum, run through the identical unreliable data as a control, was untouched. Part V.17 then closed the loop this project opened in Part V.5-6 and never fully resolved: does the classic momentum-crash mechanism, rejected there for a different signal on the full sample, actually explain why momentum broke in 2008-09? Reapplying that same rejected regression to momentum's own hedged long leg, split at the same break date, found the answer was conditional: dormant pre-2008 (statistically indistinguishable from the earlier full-sample rejection), then sharply, significantly active since. The one surviving finding in this entire guide now has not just a confirmed break date, but a named, tested reason for it. Part V.18 then took that reason abroad, to the one market this project had visited more than any other and trusted least: does the same crash mechanism show up on NSE, which lived through the same 2008 crisis? It doesn't — the specific Bear-plus-High-Vol interaction never clears significance there. But testing it required re-running NSE momentum through this project's own best hedge methodology for the first time, not the cruder regression Part V.9 had used, and that re-run found something new: a tentative signal since 2008 the earlier test had lacked the power to see. Even a market this project believed it had already closed the book on had more to say once asked with a better instrument. Part V.19 then turned that same instinct back on Part V.17's own result: was the crash mechanism's "post-2008" significance a standing regime, or one crisis pooled with eight quiet years? Splitting the window apart found the market's own trailing-bear indicator had not fired once since September 2009 — the significance Part V.17 reported had nowhere else to come from but the crisis itself, and, isolated there, even the specific interaction term stopped being what carried it. Part V.20 then asked the question Part V.19 itself invited but didn't ask: was that "never fired again" finding a fact about the market, or about the one-year window chosen to measure it? Sweeping nearby windows found both answers at once — the crash-mechanism pattern held up under most choices tested, but the specific claim that no bear market recurred did not, vanishing the moment a shorter, equally ordinary lookback was used instead. Part V.22 then closed the loop on this project's own last untested claim, aiming the same instrument at itself one more time: the tentative NSE signal Part V.18 had flagged, not as a finding, but as a "promising, not confirmed" result with a named caveat nobody had gone back to test. Splitting NSE's universe at the exact date it stopped growing found the caveat as named wasn't the problem — but the same pooled-window pattern this project had just learned to distrust in the crash mechanism was. Part V.23 then asked a question no correction could answer: not what was wrong with a finding already in hand, but whether either of this project's two markets was itself representative of anything beyond its own dataset. Finding a third, independent one took longer than running the actual test once it was found.

The practical output (Part VI) turns that into three concrete artifacts: an investment framework that explicitly forbids trusting a blend without decomposing it; a risk-management playbook built directly from a real simulated result, not a generic checklist; and a business idea whose differentiation *is* the decomposition discipline the research itself needed. Milestones 3 through 23 then ran the India findings through increasingly rigorous versions of the same skepticism the project applies to everything else, and at every step a stronger method found something the weaker one had missed or overclaimed: momentum and reversal turned out to be universe-dependent, not general truths; a value/growth confound was tested

and rejected; a formally-compelling-looking momentum-crash mechanism was tested and rejected too, once proper statistical significance testing replaced descriptive pattern-matching; the actual explanation for the losing signal, when it finally arrived, was the most basic and least glamorous of all the candidates tested — an uncontrolled beta mismatch, checked last instead of first, then confirmed a second way with an actual hedged strategy; and then, applying that same beta check to the signal the project had been treating as a validated success, the success turned out not to be one — while a signal nobody had flagged as special (US momentum) turned out to hold real, statistically robust alpha once someone finally looked properly — alpha that Part V.10 then showed had partially decayed since its 1993 publication, that Part V.11 then showed does not clear this project's own bar for a forward-sizeable edge once tested with an actual out-of-sample hedge, that Part V.12 then confirmed is genuine, ongoing decay rather than an artifact of the 2009 crash or the hedge's own design, that Part V.13 then showed is not even unique to momentum — the identical shape turned up in a signal this project had already, separately, declared dead — and that Part V.14 then showed isn't quite the shape anyone had assumed either: reversal decays the way the textbook predicts, but momentum's own decline is better described as a sudden 2008-09 break than the same slow crowding story — a claim Part V.15 then formalized and found to be simultaneously more and less true than Part V.14 suggested: real for the specific, literature-motivated date, not decisively confirmed as the single best break once the search itself was accounted for, and, for reversal, concealing a genuine break of its own decades before 2008 — a break that Part V.16 then traced to its actual cause and found was never a market phenomenon at all, retracting reversal's status as a second surviving finding entirely — while Part V.17, applying the same instinct to the finding that survived, gave momentum's own 2008-09 break a mechanism it had lacked since Part V.13 first described its shape. Part V.18 carried that mechanism to NSE and found it didn't travel — but the trip wasn't wasted: testing it exposed that NSE momentum itself had only ever been checked with this project's cruder, pre-Part-V.8 method, and the better one found a signal since 2008 that the earlier pass had missed entirely. Part V.19 then asked one question of Part V.17's own mechanism that no earlier milestone had: not whether it was real, but whether it was still happening, and found the "post-2008" window it was tested against had, itself, only one crisis in it to be significant about. Part V.20 then did to Part V.19 what Part V.19 had just done to Part V.17: checked whether its own headline finding depended on one arbitrary-looking choice. It found two different answers hiding in the same code — the crash-mechanism pattern survived most reasonable variations of that choice, but the specific claim that no bear market had recurred did not. Part V.21 then finished what Part V.20 had only made possible: it actually ran the persistence test under the windows where a second bear market exists, instead of leaving that as a flagged possibility, and found the mechanism stayed quiet — not because there was nothing to test it against this time, but because it was tested and simply didn't show up, with the one coefficient that did clear significance pointing the wrong way for crash risk entirely. Part V.22 then took the same instrument this project had just used on its own crash-mechanism finding and pointed it at the one other open thread still standing: the tentative NSE signal. It found the same failure mode again, in a completely different corner of the project — a real, named caveat that turned out not to be the actual problem, sitting next to a pooled-window artifact nobody had thought to check until the crash mechanism taught this project what to look for. Part V.23 then stepped back from correcting existing findings entirely and asked the question this project's own two-market design had left unasked since Part V.3: is either market representative of anything, or just of itself? Answering it took more effort than any single statistical test in this guide — most of the obvious candidate sources turned out to be unreachable, blocked, or not actually data at all — but the one source that worked gave momentum's long leg its cleanest replication

yet.

The fix that survived all of that scrutiny is more modest, and narrower, than any earlier draft of this conclusion claimed: there is no demonstrated, beta-independent stock-selection skill in the 52-week-high signal, in either direction, as currently built, at any point in the sample — and, as of Part V.16, the same is now true of short-term reversal, at every level this project checked. Momentum's long leg is the sole survivor: a real, largely beta-independent edge that held, essentially undiminished, all the way to the 2008-09 financial crisis, confirmed by a test built specifically around that crisis date, though a fully unconstrained search for the single best break in the series does not confirm that date as uniquely special, landing instead 27 months later without clearing significance. Reversal looked, for six milestones, like it might be a second such survivor — first retracted outright (Part V.9), then partially rehabilitated as a genuine, decayed effect (Part V.13), then given its own precise decline date (Part V.14), then its own formally-tested structural break (Part V.15) — before Part V.16 finally asked the one question none of those milestones had: was any of it real to begin with? It wasn't. A decile portfolio of two to four stocks, pulled from a universe of companies guaranteed by construction to have survived and thrived, is not evidence of a behavioral anomaly; it is a handful of individual stock histories dressed up as a diversified backtest. Momentum's own history in that same unreliable data was checked as a control and came through clean. Every one of those findings was reached not by anyone's original hypothesis about which signal should work, but by finally applying the project's own hard-won standard of rigor evenly: to a winner as well as a loser, to a signal's own more recent, more comfortable-looking result as well as its oldest one, to a signal already written off instead of assuming a full-sample null closed the question for good, to the project's own explanatory story once a result was found instead of assuming the first plausible mechanism was the right one, and finally to the data itself, checking whether a headline number rested on a real market or on three stocks and a backfilled ticker list. That is the project working as intended, including on itself: not every finding needs to be confirmed to be useful, a compelling pattern — positive or negative — is a hypothesis until it survives testing at every level of rigor available, and the single most important thread running through this entire guide is a project that kept correcting or deepening its own most recent, best-supported-looking result, twenty times in a row — six outright retractions or downward revisions, one nuanced check (Part V.10) that briefly looked like a stopping point before Part V.11 showed it wasn't, an eighth check (Part V.12) built specifically to try to explain the seventh correction away that couldn't, and a ninth (Part V.13) that went looking for the same pattern somewhere the project had stopped looking, and found it — then, rather than resting on that broader finding, ran a tenth check (Part V.14) against its own explanation for the ninth, discovered that even "two signals, one shared story" was itself an oversimplification, ran an eleventh check (Part V.15) against its own *informal* version of that discovery and found the formal, multiple-testing-corrected test was more conservative than the eyeballed comparison that had motivated it, on both signals, in different directions, ran a twelfth check (Part V.16) that did what none of the previous eleven had: it asked whether the underlying data could even support the question being asked of it, and found that for reversal, across every one of those eleven checks, it could not, ran a thirteenth check (Part V.17) that, for the first time, wasn't a correction at all: it took the one result still standing and asked not whether it was real, but why, reapplying a mechanism this project had itself rejected elsewhere and finding it fit, conditional on era, exactly where the break had been — ran a fourteenth check (Part V.18) that took that same mechanism somewhere it had never been asked before, found it didn't belong there, and, checking why, found that even the project's own earlier verdict on that market had been reached with a tool it had since outgrown, ran a fifteenth check (Part V.19) that turned the same "is it still true" question back on the mechanism itself: split apart, the "post-2008" window the

mechanism was confirmed against turned out to contain exactly one crisis and no second bear market to test persistence against, and the specific interaction term stopped being what carried the result once that one crisis was looked at on its own, ran a sixteenth check (Part V.20) that questioned the fifteenth's own instrument: was "no second bear market" a fact about the market, or about the one specific window chosen to look for one, and found it was partly the latter — the pattern survived nearby choices, the absence-claim did not, ran a seventeenth check (Part V.21) that did not just note the sixteenth check's opening but walked through it: reran the mechanism against the two windows that now had a real second bear market to test against, and found silence — not the absence of a test this time, but a test that came back negative, with the one significant coefficient it produced running in the wrong direction for crash risk altogether, and finally ran an eighteenth check (Part V.22) that took the exact instrument the seventeenth had just used — splitting a pooled window to see if a finding survived on either side of the split — and pointed it at a completely different result: this project's own tentative NSE signal, the one open thread Part V.18 had left standing with a named but never-tested caveat. The caveat, tested, wasn't the problem; the exact same pooling artifact the seventeenth check had just diagnosed elsewhere was, ran a nineteenth check (Part V.23) that wasn't aimed at any existing finding at all: it asked whether this project's entire two-market design was itself sound, went looking for an independent third market to find out, and found that the search itself — ruling out one blocked or empty source after another — took longer than the test that followed once a real one turned up, and finally ran a twentieth check (Part V.24) that finished what the nineteenth had left half-open: a third market tested on one signal only. The second and third signals, run on it, gave two different answers — one a clean confirmation, one a result that looked new until the nineteenth check's own lesson (check what a finding correlates with before counting it twice) was turned on it. Not finding an escape hatch, finding the same shape twice in independent places, discovering the two places didn't actually share a shape after all, discovering that even that discovery needed a proper test before it could be trusted, discovering that the entire multi-milestone argument about reversal had been conducted on three stocks, discovering a real reason behind the one number that survived all of it, and finally discovering that a market believed fully closed still had something left to find, are all findings: the project never found a result, an explanation, or even a way of testing an explanation, durable enough to stop re-checking. What survives is smaller and more precisely qualified than any single milestone first suggested: momentum's long leg, real and strong pre-2008-09, confirmed weakened specifically at that literature-motivated date though not provably the site of the single largest break in the series, and now understood to have broken because the classic momentum-crash mechanism — dormant for over three decades — activated during the 2008-09 crisis, is this project's one remaining, seventeen-times-checked finding, confirmed to be a US-specific mechanism after failing to replicate on NSE, confirmed on the fifteenth check for that one crisis specifically rather than for a standing post-2008 regime, confirmed on the sixteenth to be reasonably robust to nearby modeling choices while the specific claim that no second bear market has occurred since was not, and, on the seventeenth, put to the one test that claim's own qualification made possible: run against the two windows where a second bear market genuinely exists, and it did not reactivate in either. Short-term reversal is not a second one: every version of its apparent edge — the original NSE Sharpe ratio, the full-sample beta-adjusted alpha, the pre-1994 era split, the continuous decay trend, the formal structural break — traces back to the same unreliable 1972-1977 window and disappears the moment that window is excluded. One signal decays on its own schedule and by its own mechanism, sized more cautiously than the milestone that first found it would have suggested; the other was never there to begin with, once checked against data good enough to ask the question. Two threads that looked open as recently as Part V.20 are open no longer. Whether the

momentum-crash mechanism reactivated in 2011 or 2015-16, episodes this project's own default bear-market definition was too conservative to register as bear markets at all, but that two other, equally standard definitions do register: Part V.21 ran the persistence test against both of those definitions and the mechanism did not reactivate in either, with the one coefficient that did clear significance running the wrong direction for crash risk. And whether the tentative, "promising not confirmed" post-2008 NSE momentum signal — found only because this project finally ran its own best method against a market it thought it had already closed the book on — was real: Part V.22 tested the one named caveat attached to it and found a different, more fundamental problem instead, the identical pooled-window artifact Part V.21 had just taught this project to look for. And one question this project had never actually asked, because it had never had the means to: whether momentum's edge was a property of stock markets generally or of the two specific, re-exported datasets this project happened to have access to. Part V.23 went looking for a third, independent source after ruling out an entire continent's worth of blocked or empty candidates, found one in Australia's own exchange, and momentum came back significant at both daily and monthly frequency — the cleanest replication in the whole guide, on the one market tested with this project's best method from the very first pass rather than arriving at it after several corrections. Part V.24 then finished what Part V.23 had left half-done: NSE and the US mirror were both tested on all three signals from the start, ASX had only been tested on one. Reversal, run on ASX, changed nothing — the same null as everywhere else. 52-week-high did: significant hedged alpha, reopening a question Parts V.7-8 had called resolved — until checking its correlation with momentum's own ASX result (0.76-0.82) showed the two signals were picking substantially the same stocks, the same lesson Part V.23 itself had just taught this project applied to a market it had barely finished exploring. Four settled conclusions now, not three: a real, if narrower, US-specific finding for momentum, now independently replicated on a third market; two retractions, reversal's and the NSE signal's, reached for different reasons but by the same refusal to let a promising number stand without being taken apart; and a genuinely new confirmation, arrived at not by re-examining an old result but by questioning whether this project's entire evidentiary base was wide enough to trust in the first place. What remains open is not a specific finding but three honest limits of the data itself: whether the crash mechanism would reactivate in a genuinely severe future crisis, as opposed to the milder episodes this sample happens to contain, is a question no amount of further re-testing of this history can answer — the most recent decade, for momentum, to size nothing against — whether ASX momentum's own edge is stable across sub-periods the way the US finding eventually was shown to be, a check this project's newest market hasn't had enough history yet to run, and whether ASX 52-week-high carries any skill beyond what its correlation with momentum already explains, a check that needs momentum held constant as an explicit control rather than compared after the fact. All three are what a research process built to distrust its own best-looking result eventually converges on.

Glossary

Alpha

The part of a strategy's average return left over after accounting for its market exposure (beta) -- the piece that would represent genuine stock-selection skill, if it's significantly different from zero. This guide's Part V.7 finds none in the 52-week-high signal, either leg, once beta is controlled for.

Anchoring

Relying too heavily on an initial reference point (e.g. a 52-week high, an opening price) when making subsequent judgments.

Beta

How much a position tends to move for a given move in the overall market -- a beta of 1 means it moves with the market on average, above 1 means it amplifies market moves, negative means it moves opposite the market. A long-short book with unequal long and short beta carries unintended net market exposure.

Composite score

A single number blending several individual signals into one rank; this project's central finding is that a composite can hide a component that is actively harmful.

Correlation regime shift

A change in how much different assets move together — typically a jump from low/moderate to near-1 correlation during a market crisis.

Crowding

Many independent, sophisticated participants arriving at similar positions without coordinating, producing hidden correlation across supposedly diversified portfolios.

Decile / quintile backtest

A methodology that ranks a universe by a signal, then goes long the best-ranked group and short the worst-ranked group, to isolate the return spread the signal actually predicts.

Efficient Market Hypothesis (EMH)

The theory that asset prices reflect all available information, implying no strategy should reliably beat a passive index.

Extrapolation bias

Assuming a recent trend (e.g. fast earnings growth) will continue further into the future than it typically does.

Fat tails

A return distribution with more extreme outcomes than a normal (bell-curve) distribution predicts.

Herding

Following the crowd's behavior rather than independently evaluating a decision — useful most of the time, dangerous when the crowd itself started moving for an unrelated reason.

Leverage

Borrowed capital or notional derivative exposure used to scale a position beyond what the investor's own capital would allow; it magnifies both gains and losses.

Maximum drawdown

The largest peak-to-trough decline a strategy experiences at any point in a backtest.

Momentum

The tendency of a security's recent trend to continue, typically attributed to underreaction.

Momentum crash risk

The tendency of a "long recent winners, short recent losers" strategy to suffer severe losses when the short leg's high-beta losers rebound sharply, typically during high-volatility, post-downturn market recoveries.

Overconfidence

Overestimating the reliability of one's own judgment or model, often reinforced by a recent string of favorable outcomes.

Post-earnings-announcement drift (PEAD)

The tendency of a stock price to keep drifting in the direction of an earnings surprise for weeks after the announcement, rather than repricing immediately.

Newey-West (HAC) standard errors

A correction to a regression's standard errors that accounts for serial correlation in the data (e.g. from overlapping monthly holding periods); without it, significance tests on this kind of strategy-return data can look more confident than the data actually supports.

p-value

The probability of seeing a result at least this extreme if there were truly no effect. Conventionally, $p < 0.05$ is called "statistically significant" -- but with many tests run at once, some will cross that bar by chance alone (see this guide's own Milestone 5 for a worked example).

Reversal

The tendency of a very recent, sharp price move to partially revert, typically attributed to overreaction.

Sharpe ratio

Average return divided by return volatility, annualized; a standard measure of risk-adjusted performance.

Survivorship bias

The distortion introduced by only including, in a historical study, entities that are still around today (e.g. still-listed companies), which tends to overstate historical performance.

Value-at-Risk (VaR)

An estimate of how much a portfolio could lose over a given horizon at a given confidence level (e.g. the worst-case loss on all but the worst 1% of days).

Sources and Further Reading

- Roger Lowenstein, *When Genius Failed: The Rise and Fall of Long-Term Capital Management* (2000).
- President's Working Group on Financial Markets, report on hedge fund leverage and LTCM (April 1999).
- Amir Khandani & Andrew Lo, "What Happened To The Quants In August 2007?", *Journal of Investment Management*, 2011 (working paper 2007).
- Contemporaneous financial press coverage of Amaranth Advisors' September 2006 collapse (e.g. *Wall Street Journal*, September-October 2006).
- David X. Li, "On Default Correlation: A Copula Function Approach", *Journal of Fixed Income*, 2000.
- Felix Salmon, "Recipe for Disaster: The Formula That Killed Wall Street", *Wired*, February 2009.
- Michael Lewis, *The Big Short* (2010).
- US House Financial Services Committee hearing, "Game Stopped? Who Wins and Loses When Short Sellers, Social Media, and Retail Investors Collide" (February 18, 2021).
- Brad Barber & Terrance Odean, "All That Glitters: The Effect of Attention and News on the Buying Behavior of Individual and Institutional Investors", *Review of Financial Studies*, 2008.
- Narasimhan Jegadeesh & Sheridan Titman, "Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency", *Journal of Finance*, 1993.
- William George & Chuan-Yang Hwang, "The 52-Week High and Momentum Investing", *Journal of Finance*, 2004.
- Narasimhan Jegadeesh, "Evidence of Predictable Behavior of Security Returns", *Journal of Finance*, 1990.
- Josef Lakonishok, Andrei Shleifer & Robert Vishny, "Contrarian Investment, Extrapolation, and Risk", *Journal of Finance* 49(5), 1994, pp. 1541-1578.
- Richard Thaler, *Misbehaving: The Making of Behavioral Economics* (2015).
- Kent Daniel & Tobias Moskowitz, "Momentum Crashes", *Journal of Financial Economics*, 2016.
- Werner De Bondt & Richard Thaler, "Does the Stock Market Overreact?", *Journal of Finance*, 1985.
- Whitney Newey & Kenneth West, "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix", *Econometrica*, 1987 -- the standard reference for the HAC standard errors used in Part V.6.

All code, full case-study write-ups, and the underlying data-loading and backtest engine referenced throughout this guide live in the project's code repository, folder research/behavioral-finance/, which remains the definitive, most up-to-date source as milestone 3 and beyond continue.